Mathematical Simulation of Blood Flow

A Literature Review and Idealized Stenosis Study

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Abstract.

This report reviews the mathematical and numerical setting for simulating idealized stenotic blood flow and frames a controlled reduced-order stenosis study. The literature-review portion fixes the continuum, constitutive, boundary-data, and model-hierarchy vocabulary needed to move from three-dimensional (3D) Navier–Stokes and generalized-Newtonian models to reduced 0D/1D/2D and coupled hemodynamic formulations. Pressure-ratio quantities are treated as computed model outputs unless an external clinical measurement comparison is explicitly introduced. The numerical portion reports a $T = 1$ comparison between the selected 1D solver and resolved 3D velocity data for two idealized stenosis cases, using section-mean and radial-profile velocity diagnostics. The comparison shows broad trend agreement away from the stenosis but large near-throat velocity overprediction, so the record is interpreted as a diagnostic comparison rather than validation.

Keywords:

computational hemodynamics, blood flow simulation, stenotic vessels, Navier–Stokes, reduced-order models, idealized stenosis

Contents

1 Introduction and Report Scope

Coronary and arterial stenoses motivate a basic computational question because anatomical narrowing alone does not determine functional obstruction. This report is a mathematical and numerical study, not a clinical validation study: it assembles the literature and notation needed to state continuum, constitutive, boundary-data, model-hierarchy, and reduced-output conventions, then reports a first controlled Newtonian $T = 1$ velocity comparison between a selected 1D stenosis model and resolved 3D data for two idealized cases.

The report fixes the vocabulary needed to move from continuum assumptions, to the three-dimensional incompressible Navier–Stokes and generalized-Newtonian settings, to reduced 0D/1D/2D hemodynamic model classes, and finally to a selected one-dimensional numerical study of an idealized stenosis. Its contribution is twofold: it organizes the relevant mathematical and numerical literature, and it frames a controlled computation for a smooth idealized stenosed vessel.

Pressure-ratio quantities in this report are computed model outputs, not clinical FFR or CT-FFR measurements, unless an external measurement comparison is explicitly introduced. Convention C.3 in Appendix C records the pressure-reference, gauge-pressure, high-flow-window, and admissibility conventions used for any reduced pressure-ratio output. The numerical record in Section 2 reports velocity diagnostics only; it does not yet report pressure-drop or pressure-ratio outputs. Those conventions are included to fix the output vocabulary for subsequent reduced-order studies.

Report Roadmap

1. Section 1.1 fixes the continuum, kinematic, balance-law, and constitutive vocabulary used later in the report.
2. Section 1.2 identifies the governing equations, boundary-data vocabulary, and model hierarchy.
3. Section 1.3 records the pressure–flow motivation and the reduced-output conventions used to keep pressure-drop, pressure-ratio, and shear-proxy quantities separate from the velocity diagnostics demonstrated in the main numerical comparison.
4. Section 2 reports the controlled $T = 1$ comparison between the selected 1D solver and resolved 3D velocity data for two idealized stenosis cases.
5. Section 3 summarizes what the report establishes and states the limitations of the current numerical record.
6. Appendices B–H collect supporting notation, derivation details, numerical-method notes, and code-use documentation.

1.1 Continuum and Constitutive Assumptions

This section moves from the macroscopic scale assumption, to continuum field variables, to blood-dynamics kinematics, to balance-law vocabulary, and finally to Newtonian versus generalized-Newtonian stress closures.

1.1.1 Physiological Scale Boundary

The present work is posed at the macrocirculatory arterial scale. At this scale, blood is commonly modeled as a viscous fluid whose bulk behavior can be described by continuum mechanics. The goal is not to resolve the effects of individual blood cells, platelets, or plasma constituents but rather to predict macroscopic observables such as pressure, flow rate, pressure drop, and pressure ratio. [1, Sec. 1.1.2, pp. 9–10] [1, Sec. 1.1.3, pp. 12–14].

1.1.2 Blood as a Continuum

Consider a time-dependent family of spatial fluid domains. For a final time $T_{\text{final}} > 0$, use $I_T := (0, T_{\text{final}})$ for interior-time PDE statements and $\overline{I}_T := [0, T_{\text{final}}]$ for initial data and flow maps. Define

$$\begin{cases} \Omega_B(t) := \{\boldsymbol{x} \in \mathbb{R}^3 : \boldsymbol{x} \text{ lies inside the vessel at time } t\} \\ \mathcal{Q}_T := \{(t, \boldsymbol{x}) : t \in I_T,\ \boldsymbol{x} \in \Omega_B(t)\}, \end{cases}$$

where the reference configuration is denoted $\Omega_B(0)$, the current configuration is $\Omega_B(t)$, and the space-time fluid domain is $\mathcal{Q}_T$. This domain notation is the one collected in Appendix B; boundary and function-space conventions are summarized in Appendix D. In this report, the continuum assumption is a large-vessel idealization: individual blood cells, platelets, and plasma constituents are not resolved, and blood velocity, pressure, density, and Cauchy stress are represented by sufficiently regular fields on $\mathcal{Q}_T$. Cellular-scale effects may enter only through later constitutive choices [2, Part I, Ch. 1, Secs. 1.1–1.2, pp. 4–5] [1, Secs. 1.1.3 and 3.2, pp. 12–14 and 46–47]. The continuum setup is as follows: Assumption 1.1, Remark 1.2, Definition 1.3, Definitions 1.5–1.6, Definitions 1.8–1.9, and Definitions 1.13–1.14 fix the assumptions used in the model hierarchy.

1.1.3 Blood Dynamics

This subsection fixes the kinematic vocabulary used by the later continuum and reduced-order models.

Assumption 1.1 (Blood-dynamics kinematic setting)**.** For $t \in \overline{I}_T$, blood motion is represented by a sufficiently regular orientation-preserving diffeomorphism $\mathcal{L}_t : \Omega_B(0) \to \Omega_B(t)$ with $\det \nabla_{\boldsymbol{\xi}} \mathcal{L}_t > 0$. The family $\mathcal{L}_t$ has the time and space regularity needed to differentiate $\nabla_{\boldsymbol{\xi}} \mathcal{L}_t$ and $\det \nabla_{\boldsymbol{\xi}} \mathcal{L}_t$, define material derivatives, and apply Reynolds transport to material volumes $V(t) = \mathcal{L}_t(V(0))$ with Lipschitz boundary. It carries each material label $\boldsymbol{\xi}$ to its current location $\boldsymbol{x}(t, \boldsymbol{\xi}) = \mathcal{L}_t(\boldsymbol{\xi})$. The Eulerian velocity field $\boldsymbol{u} : \mathcal{Q}_T \to \mathbb{R}^3$ satisfies

$$\partial_t \mathcal{L}_t(\boldsymbol{\xi}) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi})).$$

The accompanying continuum fields are pressure $p : \mathcal{Q}_T \to \mathbb{R}$, density $\rho : \mathcal{Q}_T \to \mathbb{R}^+$, and Cauchy stress $\boldsymbol{T} : \mathcal{Q}_T \to \mathbb{R}^{3\times 3}$.

Figure 1 illustrates this Lagrangian–Eulerian notation.

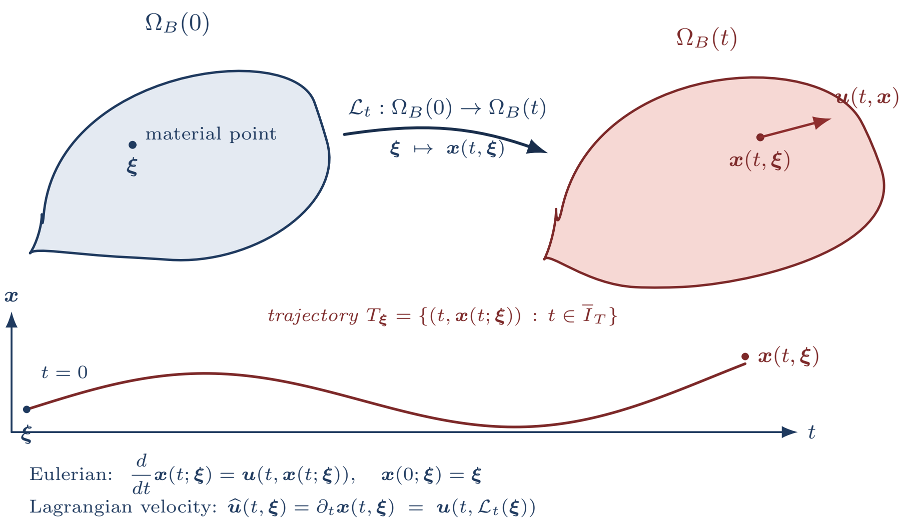

Figure 1: Lagrangian and Eulerian descriptions. The flow map $\mathcal{L}_t : \Omega_B(0) \to \Omega_B(t)$ carries the material label $\boldsymbol{\xi} \in \Omega_B(0)$ to its current position $\boldsymbol{x}(t, \boldsymbol{\xi}) \in \Omega_B(t)$. The Eulerian velocity $\boldsymbol{u}(t, \boldsymbol{x})$ is the velocity of whichever particle currently sits at $(t, \boldsymbol{x})$; the Lagrangian velocity is $\widehat{\boldsymbol{u}}(t, \boldsymbol{\xi}) = \partial_t \boldsymbol{x}(t, \boldsymbol{\xi}) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi}))$, and trajectories $T_{\boldsymbol{\xi}}$ are generated by the velocity field $\boldsymbol{u}$.

Remark 1.2 (Eulerian vs. Lagrangian). The Eulerian description views fields such as $\phi(t, \boldsymbol{x})$ or $\boldsymbol{u}(t, \boldsymbol{x})$ at fixed spatial locations. The Lagrangian description follows material particles $\boldsymbol{\xi}$ along trajectories generated by the velocity field and views fields as functions of the material label $\boldsymbol{\xi}$ and time t, such as $\widehat{\phi}(t, \boldsymbol{\xi}) = \phi(t, \mathcal{L}_t(\boldsymbol{\xi}))$ or $\widehat{\boldsymbol{u}}(t, \boldsymbol{\xi}) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi}))$. The flow map $\mathcal{L}_t$ is the change of variables between these descriptions, and the trajectory of a particle with label $\boldsymbol{\xi}$ is the characteristic curve $t \mapsto \boldsymbol{x}(t; \boldsymbol{\xi}) = \mathcal{L}_t(\boldsymbol{\xi})$. Where defined, the inverse $\mathcal{L}_t^{-1}$ maps an Eulerian location back to its material label.

Under the standard assumption that the prescribed velocity field is continuous in time and uniformly Lipschitz in space, particle trajectories exist uniquely and depend continuously on their initial labels; see [3, Ch. 2]. This is a trajectory-level ODE result and should not be confused with Navier–Stokes well-posedness.

Definition 1.3 (Material derivative)**.** For $\phi \in C^1(\mathcal{Q}_T)$, the material derivative is the derivative of ϕ along a particle trajectory:

$$D_t\phi(t,\boldsymbol{x}) \;:=\; \frac{d}{dt}\phi(t,\mathcal{L}_t(\boldsymbol{\xi})) = \partial_t\phi(t,\boldsymbol{x}) + \boldsymbol{u}(t,\boldsymbol{x})\cdot\nabla\phi(t,\boldsymbol{x}), \qquad \boldsymbol{\xi} = \mathcal{L}_t^{-1}(\boldsymbol{x}).$$

For vector fields, D_t is applied componentwise.

Let $\widehat{\boldsymbol{F}}_t = \nabla_{\boldsymbol{\xi}}\mathcal{L}_t$ be the deformation gradient and $\widehat{J}_t = \det\widehat{\boldsymbol{F}}_t$ the Jacobian determinant. In Eulerian notation this is written $J_t(\boldsymbol{x}) = \det\widehat{\boldsymbol{F}}_t(\mathcal{L}_t^{-1}(\boldsymbol{x}))$. The standard Jacobian evolution law $D_tJ_t = J_t\,\mathrm{div}\boldsymbol{u}$ is used implicitly below; derivation details are deferred to Appendix E.

Theorem 1.4 (Reynolds Transport Theorem)**.** *Let $V(t) = \mathcal{L}_t(V(0))$ be a bounded material volume with Lipschitz boundary. If $\phi\in C^1(\mathcal{Q}_T)$, then*

$$\begin{aligned}\frac{d}{dt}\int_{V(t)}\phi\;d\boldsymbol{x} &= \int_{V(t)}(D_t\phi+\phi\;div\boldsymbol{u})\;d\boldsymbol{x}\\ &= \int_{V(t)}(\partial_t\phi+div(\phi\boldsymbol{u}))\;d\boldsymbol{x}.\end{aligned}$$

Proof-level details for Theorem 1.4 and the momentum balance below are found in Appendix E.

Continuum assumptions retained in the main model

Assumption 1.1 supplies the 3D velocity, pressure, density, and stress fields used below. The balance-law vocabulary developed below permits variable density. The reduced arterial models considered later adopt the constant-density specialization $\rho\equiv\rho_0>0$, which is standard for large-vessel incompressible blood-flow models.

The domains and material control volumes are assumed regular enough for traces, outward unit normals, and the divergence theorem; Lipschitz boundaries are sufficient for the integral manipulations used here; see Convention D.1 in Appendix D.

Definition 1.5 (Material density preservation)**.** A flow has material density preservation if density is preserved along particle trajectories:

$$D_t\rho = 0.$$

Definition 1.6 (Incompressible flow)**.** A flow is incompressible, or volume preserving, if material volumes are preserved. With $\mathcal{L}_0 = \mathrm{Id}$ this is equivalent to

$$J_t\equiv 1 \quad\Longleftrightarrow\quad \mathrm{div}\boldsymbol{u} = 0.$$

Remark 1.7 (Divergence-free condition)*.* Mass conservation gives

$$\partial_t \rho + \operatorname{div}(\rho \boldsymbol{u}) = 0 \quad \iff \quad D_t \rho + \rho \operatorname{div} \boldsymbol{u} = 0.$$

In the constant-density model used for incompressible Navier–Stokes, $\rho \equiv \rho_0 > 0$, so mass conservation reduces to $\operatorname{div}\boldsymbol{u} = 0$ in $\mathcal{Q}_T$.

Definition 1.8 (Linear momentum balance)**.** For a material fluid element $V(t) \subset \Omega_B(t)$, the linear momentum balance is

$$\frac{d}{dt} \int_{V(t)} \rho \boldsymbol{u} \, dV = \int_{\partial V(t)} \boldsymbol{T} \hat{\boldsymbol{n}} \, dS + \int_{V(t)} \rho \boldsymbol{f_b} \, dV,$$

where $\boldsymbol{T}$ is the Cauchy stress tensor and $\boldsymbol{f_b}$ is body force per unit mass.

Definition 1.9 (Strong momentum balance)**.** Under the regularity and mass-balance assumptions above, the local momentum balance is

$$\rho \left(\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla) \boldsymbol{u} \right) = \mathbf{div}(\boldsymbol{T}) + \rho \boldsymbol{f_b}, \qquad \text{in } \Omega_B(t).$$

The stress law closes the momentum equation. Let the rate-of-deformation tensor be

$$\boldsymbol{D}(\boldsymbol{u}) \; := \; \frac{1}{2} \left(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top \right).$$

Definition 1.10. A fluid is Newtonian if, after separating the pressure contribution, its viscous stress depends linearly on the rate-of-deformation tensor $\boldsymbol{D}(\boldsymbol{u})$.

Remark 1.11 (Blood isotropy justification)*.* Blood is often modeled as an isotropic continuum fluid at large-vessel scales. This is a closure assumption for averaged behavior, not a microscopic statement about cells suspended in plasma. The cited sources support the scale and rheology caveat–nearly constant large-vessel relative viscosity versus diameter- and hematocrit-dependent apparent viscosity in small tubes–not a direct cellular-level isotropy claim [1, Sec. 1.1.3, Fig. 1.14, pp. 12–14, Sec. 3.2, pp. 46–47] [4, Abstract, pp. H1770–H1772].

Remark 1.12 (Isotropic constitutive response)*.* An isotropic fluid has a constitutive response that is independent of the chosen coordinate system. Consequently, the Cauchy stress depends on coordinate-free invariants of $\boldsymbol{D}(\boldsymbol{u})$ rather than on a particular basis.

Definition 1.13 (Newtonian, isotropic constitutive law)**.** For a Newtonian, isotropic fluid,

$$\boldsymbol{T} = -p \, \mathbf{I} + 2\eta \, \boldsymbol{D}(\boldsymbol{u}) + \lambda \operatorname{div}(\boldsymbol{u}) \, \mathbf{I},$$

where η is the dynamic viscosity and λ is the coefficient of the volumetric-viscous term.

Definition 1.14 (Incompressible stress tensor)**.** When $\operatorname{div}\boldsymbol{u} = 0$, the Newtonian incompressible Cauchy stress is

$$\boldsymbol{T} = -p\,\mathbf{I} + 2\eta\,\boldsymbol{D}(\boldsymbol{u}).$$

More general models allow viscosity to vary with shear rate. For generalized Newtonian blood models, the dynamic viscosity is allowed to depend on the local deformation-rate magnitude [2, Part I, Ch. 1, Secs. 2.7–3.1, pp. 16–19]. A standard constitutive form is

$$\dot\gamma \;:=\; \sqrt{2\boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u})},$$

$$\boldsymbol{T} = -p\,\mathbf{I} + 2\eta_{\text{eff}}(\dot\gamma)\boldsymbol{D}(\boldsymbol{u}).$$

Here

$$\|\boldsymbol{D}(\boldsymbol{u})\|^2 \;:=\; \boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u}) = \sum_{i,j} D_{ij}(\boldsymbol{u})^2$$

is the pointwise Frobenius norm squared. When a source writes the viscosity as a function of $D : D$ or $|D|_{\mathrm{F}}^2$, this report translates it through the scalar shear-rate convention above. Here η_{eff} is an effective dynamic viscosity. Kinematic viscosity is $\nu = \eta/\rho$ after division by density. The Newtonian incompressible model is recovered from this generalized Newtonian form when η_{eff} is constant.

Definition 1.15 (Rheology closures used by the numerical record)**.** The Julia implementation exposes the rheology descriptor $\mathcal{R}$, with admitted values `newtonian`, `carreau`, `carreau-yasuda`, `casson`, and `power-law`. The descriptor selects a scalar effective dynamic viscosity law $\eta_{\text{eff}}^{\mathcal{R}}(\dot\gamma)$. For computed records, all non-Newtonian laws are evaluated at the positive shear-rate argument

$$\dot\gamma_\epsilon \;:=\; \max\{|\dot\gamma|, \epsilon_\gamma\}, \qquad \epsilon_\gamma > 0, \qquad [\epsilon_\gamma] = \mathrm{s}^{-1}.$$

The same floor is used for Casson and power-law records, where the formulas are singular or degenerate at zero shear for common parameter ranges. Optional dynamic-viscosity clamps satisfy $0 \le \eta_{\min} \le \eta_{\max} \le \infty$

with $\eta_{\max} > 0$; the clamp is applied after evaluating the displayed model law.

$$
\begin{aligned}
\eta_{\text{eff}}^{\text{newt}} &\equiv \eta, \\
\eta_{\text{eff}}^{\text{carreau}}(\dot{\gamma}) &= \eta_\infty + (\eta_0 - \eta_\infty)\left(1 + (\lambda_{\text{C}}\dot{\gamma}_\epsilon)^2\right)^{(n_{\text{C}}-1)/2}, \\
\eta_{\text{eff}}^{\text{CY}}(\dot{\gamma}) &= \eta_\infty + (\eta_0 - \eta_\infty)\left(1 + (\lambda_{\text{CY}}\dot{\gamma}_\epsilon)^{a_{\text{Y}}}\right)^{(n_{\text{CY}}-1)/a_{\text{Y}}}, \\
\eta_{\text{eff}}^{\text{casson}}(\dot{\gamma}) &= \left(\sqrt{\tau_y/\dot{\gamma}_\epsilon} + \sqrt{\eta_p}\right)^2, \\
\eta_{\text{eff}}^{\text{power}}(\dot{\gamma}) &= K_{\text{pl}}\dot{\gamma}_\epsilon^{n_{\text{pl}}-1}.
\end{aligned}
$$

The parameter domains are

$$
\eta_0 > 0, \quad \eta_\infty \geq 0, \quad \eta_0 \geq \eta_\infty, \quad \lambda_{\text{C}}, \lambda_{\text{CY}} \geq 0, \quad n_{\text{C}}, n_{\text{CY}} > 0, \quad a_{\text{Y}} > 0,
$$

for the Carreau and Carreau–Yasuda families,

$$
\tau_y \geq 0, \qquad \eta_p > 0
$$

for Casson, and

$$
K_{\text{pl}} > 0, \qquad n_{\text{pl}} > 0
$$

for the power-law closure. Under the SI convention, η, η_0, η_∞, η_p, $\eta_{\min}$, and $\eta_{\max}$ have units $\text{Pa} \cdot \text{s}$, λ_{C} and λ_{CY} have units s, τ_y has units Pa, and K_{pl} has units $\text{Pa} \cdot \text{s}^{n_{\text{pl}}}$. In the active cgs numerical records of Convention C.2, the corresponding dynamic-viscosity parameters are in g/(cm s), τ_y is in dyn/cm^2, and K_{pl} is in $\text{dyn}\,\text{s}^{n_{\text{pl}}}/\text{cm}^2$. The Yasuda transition exponent is denoted a_{Y} here so that it is not confused with the solver coordinate $a = R^2$ used in the reduced numerical record. These are pointwise generalized-Newtonian closures for the viscosity coefficient; the reduced 1D implementation uses them as an effective viscosity hook, not as a new closure-specific 1D derivation [2, Part I, Ch. 1, Secs. 2.7–3.1, pp. 16–19] [5, 6].

Remark 1.16 (Newtonian Blood Justification). For large arteries, empirical studies suggest that apparent viscosity is often approximately constant across the relevant operating range, motivating the Newtonian approximation in many reduced-order hemodynamic models. However, diameter-dependent, hematocrit-dependent, and shear-dependent effects become increasingly important in smaller vessels and low-shear regions [1, Sec. 1.1.3, pp. 12–14] [4].

Together, Assumption 1.1 and the balance-law and stress-closure definitions in this subsection form the continuum contract used by the following hierarchy: material volumes and Reynolds transport support the mass and momentum balances, while incompressibility and the selected constitutive law specialize those balances into the PDE tiers that are later reduced by cross-sectional averaging.

1.2 Governing Equations and Model Hierarchy

The continuum assumptions of Section 1.1 lead to a layered modeling hierarchy. In particular, Assumption 1.1 supplies the flow map, material volumes, and continuum fields on which the conservation laws act. The high-fidelity starting point is the incompressible Navier–Stokes or generalized-Newtonian system on a three-dimensional lumen. The report then uses cross-sectional reduction to identify the selected 1D area-flow variables and to state the closure assumptions that must be fixed before any reduced numerical study is interpreted.

1.2.1 Conservation Laws and Navier–Stokes Specialization

The governing equations originate from conservation of mass and linear momentum applied to material volumes under Assumption 1.1, followed by the constant-density incompressible specialization. The derivation details and row-divergence convention are recorded in Appendix E and Definition D.6. Figure 2 summarizes the route from these balance laws to the forms used in the report [2, Part I, Ch. 1, Secs. 2.2–3.1, pp. 11–19, Part IV, Ch. 1, Sec. 1.1, pp. 380–383] [7, Sec. 6].

With this kinematic setting fixed, mass conservation for the constant-density incompressible model gives

$$\mathrm{div}\boldsymbol{u} = 0 \qquad \text{in } \Omega_B(t).$$

Linear momentum conservation balances inertia, pressure, viscous stress, and body forces:

$$\rho\left(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) \;=\; \mathbf{div}(\boldsymbol{T}) + \rho \boldsymbol{f_b}.$$

Throughout the Navier–Stokes statements below, $\boldsymbol{f_b}$ denotes body force per unit mass, so $\rho\boldsymbol{f_b}$ is the corresponding body force per unit volume.

Definition 1.17 (Conservative incompressible Navier–Stokes)**.** On $\Omega_B(t)$, the constant-density Newtonian system may be written in conservative tensor-divergence form as

$$\begin{cases} \partial_t(\rho\boldsymbol{u}) + \mathbf{div}(\rho\boldsymbol{u}\otimes\boldsymbol{u}) = -\nabla p + \mathbf{div}(2\eta\boldsymbol{D}(\boldsymbol{u})) + \rho\boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0, \qquad \rho \equiv \rho_0 > 0. \end{cases}$$

For constant density and divergence-free velocity, $\mathbf{div}(\rho\boldsymbol{u}\otimes\boldsymbol{u}) = \rho(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$. Thus the same Newtonian momentum equation can be read in advective form.

Definition 1.18 (Generalized-Newtonian incompressible Navier–Stokes)**.** For an incompressible generalized-

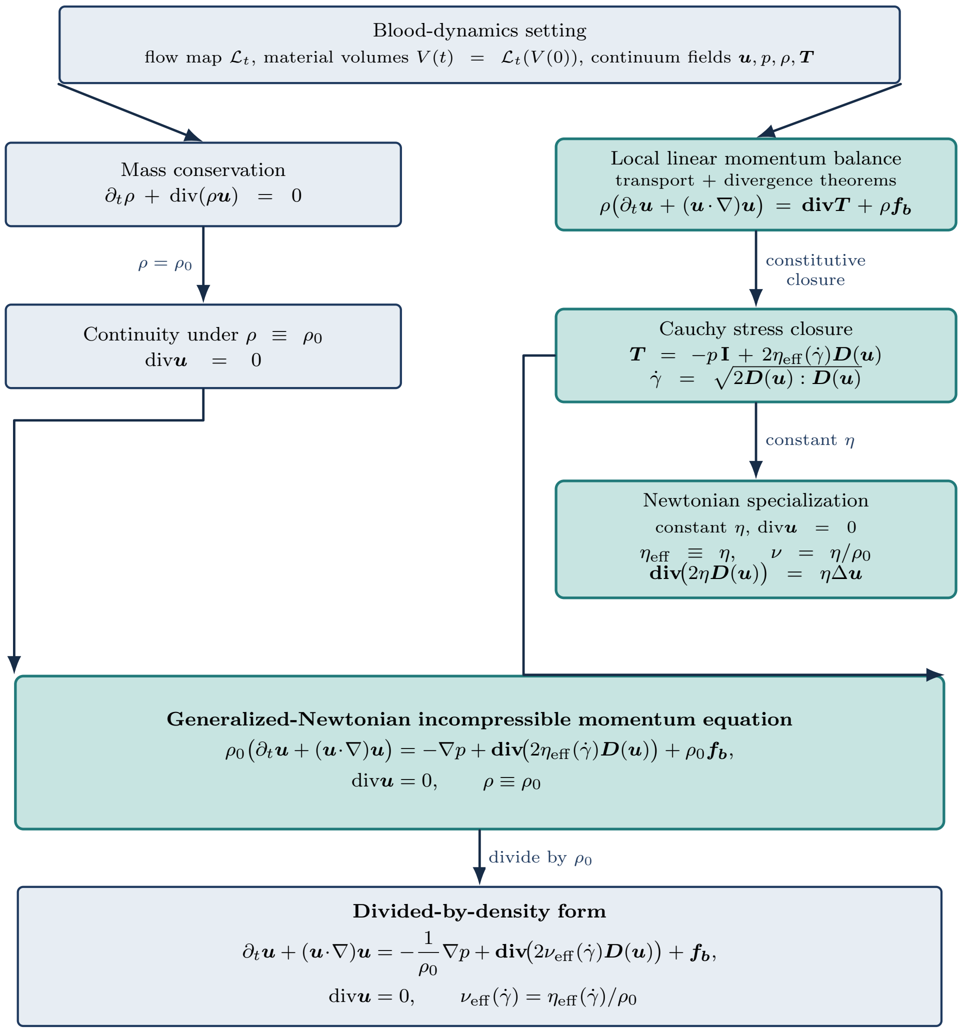

Figure 2: From the blood-dynamics setting to incompressible Navier–Stokes. The flow map and material-volume assumptions support the conservation statements. Mass conservation reduces to $\operatorname{div}\boldsymbol{u} = 0$ under constant density. Momentum balance closes through the Cauchy stress law; the Newtonian limit uses constant dynamic viscosity, while the generalized-Newtonian form keeps $\eta_{\text{eff}}(\dot\gamma)$ explicit.

Newtonian fluid, use the shear-rate convention

$$\dot\gamma \;:=\; \sqrt{2\boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u})}.$$

The velocity $\boldsymbol{u}$, pressure p, body force per unit mass $\boldsymbol{f_b}$, and constant density $\rho \equiv \rho_0 > 0$ satisfy

$$\begin{cases} \rho\big(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\big) = -\nabla p + \mathbf{div}\Big(2\eta_{\text{eff}}(\dot\gamma)\boldsymbol{D}(\boldsymbol{u})\Big) + \rho \boldsymbol{f_b}, \\ \operatorname{div}\boldsymbol{u} = 0. \end{cases}$$

Definition 1.19 (Density-scaled generalized Navier–Stokes)**.** With $\nu_{\text{eff}}(\dot\gamma) \;:=\; \eta_{\text{eff}}(\dot\gamma)/\rho$, the divided-by-density form is

$$\begin{cases} \partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\rho^{-1}\nabla p + \mathbf{div}\Big(2\nu_{\mathrm{eff}}(\dot\gamma)\boldsymbol{D}(\boldsymbol{u})\Big) + \boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0. \end{cases}$$

The density-scaled form is particularly useful for numerical simulations, because it separates the convective ($(\boldsymbol{u} \cdot \nabla)\boldsymbol{u}$) and diffusive / viscous-stress ($\mathbf{div}(2\nu_{\mathrm{eff}}(\dot\gamma)\boldsymbol{D}(\boldsymbol{u}))$) contributions in the momentum equation, allowing for more straightforward numerical treatment of each term independently.

Definition 1.20 (Newtonian incompressible Navier–Stokes)**.** If $\eta_{\mathrm{eff}} \equiv \eta$ is constant, then sufficiently smooth divergence-free velocity fields satisfy $\mathbf{div}(2\eta\boldsymbol{D}(\boldsymbol{u})) = \eta\Delta\boldsymbol{u}$. The Newtonian equations are

$$\begin{cases} \rho\big(\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u}\big) = -\nabla p + \eta\Delta\boldsymbol{u} + \rho\boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0, \end{cases} \qquad \begin{cases} \partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\rho^{-1}\nabla p + \nu\Delta\boldsymbol{u} + \boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0, \end{cases}$$

where $\nu := \eta/\rho$.

Thus the 3D PDE tier is not introduced independently: flow maps and material volumes support Reynolds transport, Reynolds transport gives mass and momentum balance, the Cauchy stress law closes momentum, and incompressibility selects the Navier–Stokes systems above.

Definition 1.21 (Navier–Stokes residual notation)**.** The density-scaled generalized system can be summarized by a momentum residual operator:

$$\begin{cases} \mathcal{R}_{\mathrm{NS}}(\partial_t \boldsymbol{u}, \nabla\boldsymbol{u}, \nabla p, \boldsymbol{u}, p; \rho, \nu_{\mathrm{eff}}) = \boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0. \end{cases}$$

Remark 1.22 (Navier–Stokes system classification). Navier–Stokes is a nonlinear coupled PDE system. The unknowns are the three velocity components and pressure. The convective term transports momentum, the pressure gradient enforces incompressibility, and the viscous term diffuses momentum through the stress tensor.

1.2.2 Boundary Data and Well-Posedness Context

For a single-vessel three-dimensional domain, write the boundary decomposition

$$\partial\Omega_B(t) = \Gamma_{\mathrm{in}}(t) \cup \Gamma_{\mathrm{out}}(t) \cup \Gamma_w(t).$$

Here $\Gamma_{\mathrm{in}}(t)$ denotes an inlet portion where a velocity, flow-rate, or profile condition may be prescribed. The outlet portion $\Gamma_{\mathrm{out}}(t)$ carries pressure, traction, or reduced downstream network data. The wall portion

$\Gamma_w(t)$ carries either a no-slip or no-penetration condition for a fixed-wall continuum model, or a wall-law or structure closure when wall motion is retained. A complete initial-boundary value problem also states $\boldsymbol{u}(0,\cdot) = \boldsymbol{u}_0$, the domain, forcing, compatibility assumptions, and the solution class.

Remark 1.23 (Global regularity context). For smooth, spatially localized initial data in three dimensions, global existence and smoothness for Navier–Stokes remains an open Clay Millennium Prize problem [8, Problem statement]. This is mathematical context for the 3D continuum model, not a direct limitation of the controlled reduced 1D solver introduced below.

Well-posedness results for a three-dimensional Navier–Stokes initial-boundary value problem are problem-specific: the statement must fix the domain class, boundary conditions, forcing, compatibility assumptions, initial-data space, solution class, and time horizon. The present report uses that vocabulary only as PDE background for the model hierarchy. It does not use the Clay global-regularity problem as support for a local theorem, and it does not prove a well-posedness result for the moving arterial domain or for the reduced numerical solver. The domain and function-space notation relevant to such statements is summarized in Appendix D; coordinate and energy background is collected in Appendix F.

1.2.3 Cross-Sectional Reduction and Selected 1D Model

The 1D model reduces the spatial dimension by assuming that the vessel has a distinguished axial direction and that pressure and velocity can be averaged over cross-sections. This keeps pulse-wave and pressure-drop structure while discarding secondary flow, separation, recirculation, and resolved wall traction. The reduction follows standard compliant-artery and dimension-reduced modeling arguments [9, Sec. 2, pp. 253–257] [1, Ch. 3, Secs. 3.2 and 3.6] [10, Sec. 3, pp. 24–26]. The reduced variables A, Q, $\bar{u}$, and p_{g} should therefore be read as cross-sectional outputs of the continuum fields under additional averaging, geometry, wall-law, and profile assumptions, not as independent replacements for the blood-dynamics setting above.

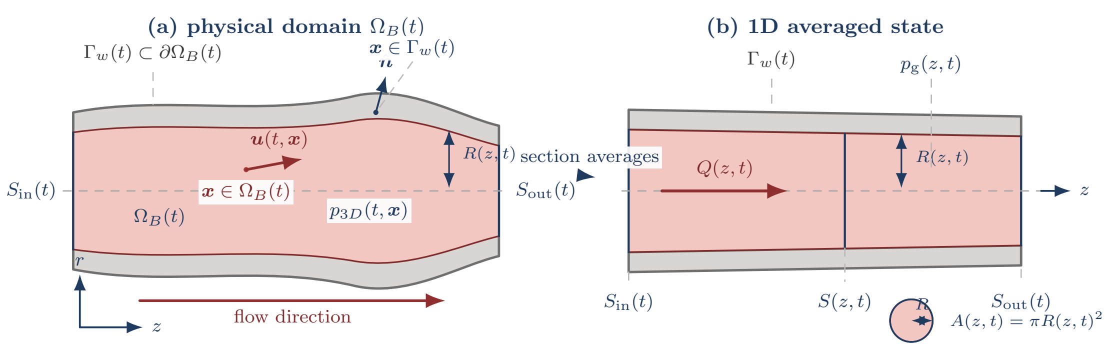

Figure 3: Compliant-vessel notation for the 1D reduction. The physical lumen $\Omega_B(t)$ is represented by a straightened averaging domain with cross-section $S(z,t)$, area $A(z,t)$, flow $Q(z,t)$, radius $R(z,t)$, and mean gauge pressure $p_{\mathrm{g}}(z,t)$.

Assumption 1.24 (Straight single-vessel geometry)**.** The selected 1D lane assumes a vessel of length L

aligned with the z-axis and a circular cross-section with radius $R(z,t)$. For $0 < z < L$, $S(z,t)$ is the interior cross-section used for averaging, not a boundary component of the open lumen.

Definition 1.25 (Section-averaged 1D fields)**.** The retained reduced variables are

$$\begin{aligned} A(z,t) &:= \int_{S(z,t)} 1\, dS, && \text{cross-sectional area,} \\ Q(z,t) &:= \int_{S(z,t)} u_z\, dS, && \text{volumetric flow rate,} \\ \bar{u}(z,t) &:= \frac{Q(z,t)}{A(z,t)}, && \text{mean axial velocity,} \\ p_{\mathrm{g}}(z,t) &:= \frac{1}{A(z,t)} \int_{S(z,t)} p_{3D,\mathrm{g}}\, dS, && \text{mean gauge pressure.} \end{aligned}$$

Here $p_{3D,\mathrm{g}}$ is continuum pressure in the wall-law gauge frame. The pressure-reference convention for reduced ratios is fixed separately in Convention C.3.

Definition 1.26 (Wall and FSI closure catalog)**.** The descriptor $\mathcal{W}$ records how the wall or wall-reduced mechanics are closed. It is independent of the rheology descriptor $\mathcal{R}$: $\mathcal{R}$ changes the stress or effective-viscosity law, while $\mathcal{W}$ changes whether the vessel wall is fixed, algebraically reduced, coupled to another model tier, or resolved as a structure.

Closure descriptor	**Wall information retained**	**Reading in this report**
$\mathcal{W}_{\mathrm{rigid}}$	Fixed or prescribed wall geometry; no structural state is solved	A fixed-wall continuum or rigid-tube reduced baseline. This is not an FSI model, even when no-slip or no-penetration wall data are imposed.
$\mathcal{W}_{\mathrm{elastic1D}}$	Dynamic area $A(z,t)$, gauge pressure $p_{\mathrm{g}}(A,z,t)$, reference area $A_0(z)$, wall stiffness $\beta(z)$, and external pressure	The selected single-vessel model. Wall mechanics are reduced to a pressure–area law, wave speed, elastic potential, and geometry source; no separate structural PDE is solved.
$\mathcal{W}_{\mathrm{multiscale}}$	Interface pressure, flow, traction, or characteristic data coupling 0D, 1D, and 3D model tiers	A boundary or coupling model requiring a separate interface and stability contract; it is not implied by the single-vessel flux/source notation.
$\mathcal{W}_{\mathrm{resolvedFSI}}$	Fluid velocity and pressure plus wall displacement, wall velocity, structural stress, and interface conditions	A full moving-wall FSI PDE system. It is a comparison tier for this report, not the active model in the first $T = 1$ study.

This separation follows the standard distinction between reduced compliant vessel models, geometrical multiscale coupling, and resolved FSI models [9, Sec. 2, pp. 253–257] [11] [12].

Definition 1.27 (Closed 1D area-flow model)**.** For $\mathcal{W} = \mathcal{W}_{\mathrm{elastic1D}}$ and an incompressible fluid with density

ρ, the working compliant single-vessel model is written with the composite axial pressure derivative

$$\frac{d}{dz}p_{\mathrm{g}}(A(z,t),z,t) = \partial_A p_{\mathrm{g}}(A,z,t)\partial_z A + \left.\partial_z p_{\mathrm{g}}(A,z,t)\right|_A .$$

The reduced balance law is

$$\partial_t A + \partial_z Q = 0, \tag{1.1}$$

$$\partial_t Q + \partial_z\left(\alpha\frac{Q^2}{A}\right) + \frac{A}{\rho}\frac{d}{dz}p_{\mathrm{g}}(A(z,t),z,t) = -C_f(z,A,Q)\frac{Q}{A}. \tag{1.2}$$

The parameter α is the momentum-flux correction factor induced by the selected cross-sectional velocity profile, C_f is the wall-friction closure, and $p_{\mathrm{g}}(A,z,t)$ is the wall-law gauge pressure closing the reduced system. This is the reduced $\mathcal{W}_{\mathrm{elastic1D}}$ member of the FSI closure axis: wall response is represented through the pressure–area law rather than through a separate structural PDE [9, Sec. 2, pp. 253–255] [1, Ch. 3, Sec. 3.2, pp. 40–47].

Assumption 1.28 (Velocity-profile and rheology closure)**.** The reduced numerical lane records a velocity-profile descriptor $\mathcal{P}$. The default is the parabolic Poiseuille profile, for which $\alpha_{\mathcal{P}} = 4/3$. A flat profile uses $\alpha_{\mathcal{P}} = 1$ for momentum transport together with a separately recorded finite shear factor. More generally, the implemented power-family profile is

$$u_{\mathcal{P}}(r) = \frac{\gamma+2}{\gamma}\,\bar{u}\left(1-\left(\frac{r}{R}\right)^{\gamma}\right), \qquad \alpha_{\mathcal{P}} = \frac{\gamma+2}{\gamma+1}, \qquad g_{\mathcal{P}} = \gamma+2.$$

Thus $\gamma = 2$ gives the parabolic case and the former $\alpha = 1.1$ record corresponds to $\gamma = 9$. The scalar constant-viscosity specialization may be read through the compact Poiseuille-calibrated wall-friction coefficient

$$C_f := \frac{8\pi\eta}{\rho}.$$

For non-Newtonian descriptor $\mathcal{R}$, the current implementation keeps the same finite-volume equation structure and evaluates a local effective kinematic viscosity

$$\dot{\gamma}_{1D} := g_{\mathcal{P}}\frac{|Q|}{a\sqrt{a}},$$
$$\nu_{\mathrm{eff}}^{\mathcal{R}} := \frac{\eta_{\mathrm{eff}}^{\mathcal{R}}(\dot{\gamma}_{1D})}{\rho},$$

using the implementation coordinate $a = R^2$. This is a numerical contract for controlled comparison records: $\nu_{\mathrm{eff}}^{\mathcal{R}}$ replaces the former scalar ν in the existing reduced viscous/source and pressure-correction terms only. It should not be read as a claim that the flux convention, friction law, rheology hook, and derived shear proxy form a complete resolved velocity-profile model.

Definition 1.29 (Elastic pressure–area wall law)**.** The baseline pressure closure is the elastic wall law

$$p_{\mathrm{g}}(A,z,t)=p_{\mathrm{ext}}(z,t)+\frac{\beta(z)}{A_0(z)}\left(\sqrt{A(z,t)}-\sqrt{A_0(z)}\right). \tag{1.3}$$

Here $A_0(z)$ is the reference area, p_{ext} is the external pressure, and $\beta(z)$ is a wall-stiffness parameter. Some implementations absorb A_0^{-1} into β; the essential closure is that p_{g} is known once A_0, β, and p_{ext} are supplied. Resolved FSI would replace this algebraic pressure–area closure with structural unknowns and fluid–structure interface conditions.

Definition 1.30 (Stenosis reference geometry)**.** A stenosis is encoded through the reference radius or reference area. Let $R_{\mathrm{base}}(z)$ be the nominal healthy radius and $s(z)\in[0,1)$ a fractional radius-reduction profile. Then

$$R_0(z)=R_{\mathrm{base}}(z)\big(1-s(z)\big), \qquad\qquad A_0(z)=\pi R_0(z)^2. \tag{1.4}$$

The baseline idealized stenotic-vessel profile is the smooth asymmetric kernel

$$g(z)=z-z_s+\kappa\exp\left(-\frac{(z-z_a)^2}{2\sigma_a^2}\right), \qquad\qquad \psi(z)=\exp\left(-\lambda g(z)^4\right), \tag{1.5}$$

$$s(z)=s_{\max}\psi(z), \qquad 0\le s_{\max}<1. \tag{1.6}$$

In the default centimeter-scale simulation record, $z_a=2.5\,\mathrm{cm}$, $z_s=3.4\,\mathrm{cm}$, $\sigma_a=1\,\mathrm{cm}$, $\kappa=0.95\,\mathrm{cm}$, and $\lambda=50\,\mathrm{cm}^{-4}$. The geometry is an idealized numerical design choice, not a patient-specific plaque morphology or clinical calibration.

Remark 1.31 (Stenosis-profile and variable-radius boundary). The profile in Eq. 1.6 is C^∞ and rapidly localized rather than compactly supported. This smooth idealized geometry is the baseline for the reduced simulations in this report. Canic et al. show that standard 1D variable-geometry models may miss stenotic pressure and velocity trends unless additional variable-radius terms are included [13, Abstract, Secs. 1–2]. Those corrections motivate later comparison questions, but they are not active in the baseline model unless a separate closure identifier and numerical contract are declared.

Definition 1.32 (Conservative flux/source notation for the reduced wall law)**.** For a reduced wall closure

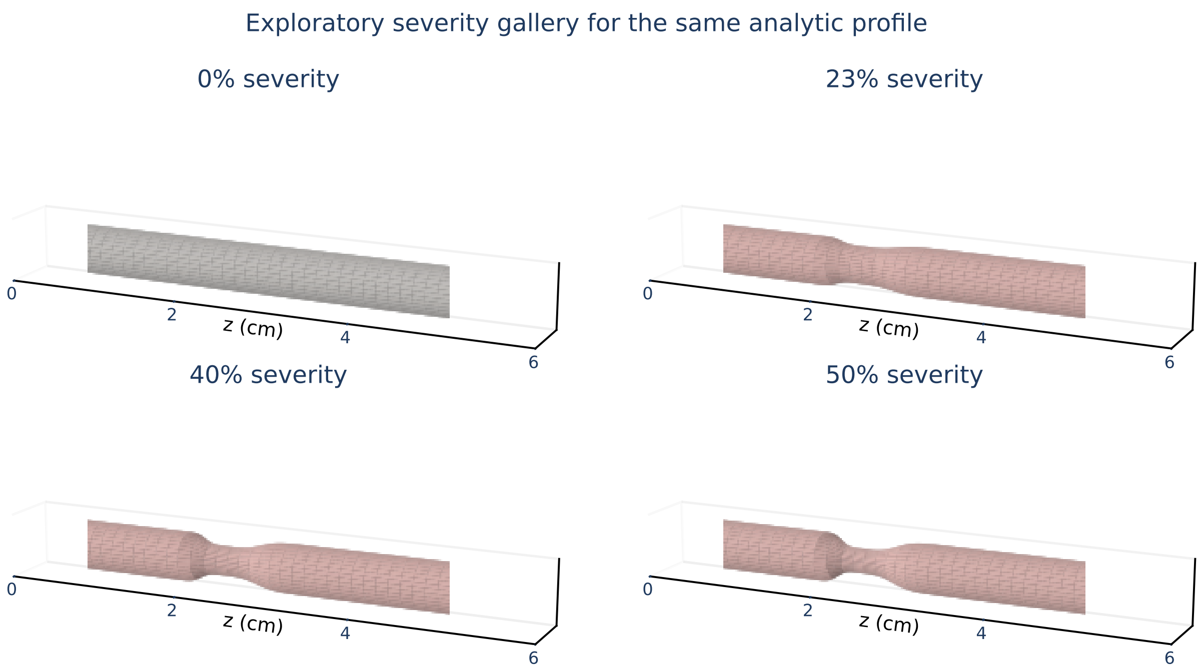

Figure 4: Analytic three-dimensional renderings of the C^∞ reference geometry in Definition 1.30, shown for $s_{\max} = 0\%, 23\%, 40\%, 50\%$. Here $s_{\max}$ is the prescribed anatomical radius-reduction parameter in the idealized geometry. These renderings illustrate the geometry family only; they are not pressure-drop, FFR, or patient-specific clinical calibration measurements.

$\mathcal{W}$ admitting an elastic potential, define

$$\mathbf{U} = (A, Q)^\top, \qquad \mathbf{F}_{\mathcal{W}}(\mathbf{U}, z, t) = \begin{pmatrix} Q \\ \alpha Q^2/A + \Psi_{\mathcal{W}}(A, z, t) \end{pmatrix},$$

$$\mathbf{S}_{\mathcal{W},\mathcal{R}}(\mathbf{U}, z, t) = \begin{pmatrix} 0 \\ S^{\mathcal{W}}_{\text{geom}}(A, z, t) + S^{\mathcal{R}}_f(A, Q, z) \end{pmatrix}.$$

The potential and sources are tied to the selected closure descriptors by

$$\begin{aligned} \partial_A \Psi_{\mathcal{W}}(A, z, t) &= \frac{A}{\rho} \partial_A p^{\mathcal{W}}_{\text{g}}(A, z, t), \\ S^{\mathcal{W}}_{\text{geom}}(A, z, t) &= \partial_z \Psi_{\mathcal{W}}(A, z, t)\big|_A - \frac{A}{\rho}\, \partial_z p^{\mathcal{W}}_{\text{g}}(A, z, t)\big|_A, \\ S^{\mathcal{R}}_f(A, Q, z) &= -C^{\mathcal{R}}_f(z, A, Q)\frac{Q}{A}. \end{aligned}$$

For $\mathcal{W}_{\mathrm{elastic1D}}$, $p_{\mathrm{g}}^{\mathcal{W}}$ is the wall law in Definition 1.29. In component form,

$$\partial_t A + \partial_z Q = 0, \tag{1.7}$$

$$\partial_t Q + \partial_z \left(\alpha \frac{Q^2}{A} + \Psi_{\mathcal{W}}(A, z, t) \right) = S_{\mathrm{geom}}^{\mathcal{W}}(A, z, t) + S_f^{\mathcal{R}}(A, Q, z), \tag{1.8}$$

The wall/FSI descriptor $\mathcal{W}$ determines the pressure–area law, elastic potential, wave speed, and geometry source; the rheology descriptor $\mathcal{R}$ determines the viscosity or friction hook. A resolved FSI model is therefore not obtained by merely relabeling S_{geom}: it introduces structural state and interface laws. The displayed split records the balance-law objects needed by finite-volume schemes, while a computed realization may still use a particular well-balanced or source-split discretization [14] [15, Summary, Secs. 2.1–2.3, Secs. 5–6] [12].

Definition 1.33 (Single-vessel boundary data)**.** For a single-vessel solve on $z \in [0, L]$, prescribe one inlet condition and one outlet condition:

$$\begin{aligned} Q(0,t) &= Q_{\mathrm{in}}(t) \quad \text{or} \quad \bar{u}(0,t) = \bar{u}_{\mathrm{in}}(t), \\ p_{\mathrm{g}}(L,t) &= p_{\mathrm{out,g}}(t) \quad \text{or} \quad p_{\mathrm{g}}(L,t) = p_{\mathrm{WK,g}}(Q(L,\cdot),t). \end{aligned}$$

The second outlet option denotes a terminal Windkessel-style relation. The first controlled study uses prescribed $Q_{\mathrm{in}}(t)$ and $p_{\mathrm{out,g}}(t)$ because they give an unambiguous reduced single-vessel contract.

Assumption 1.34 (Subcritical boundary regime)**.** The baseline boundary treatment is used only in the subcritical regime $\lambda_- < 0 < \lambda_+$, where one characteristic enters through the inlet and one characteristic enters through the outlet.

Under this regime, a finite-volume scheme constrains the incoming information at each boundary and leaves the outgoing state determined by the interior solve. This is the reduced-model analogue of declaring enough boundary data for the selected equation class; it is not, by itself, a validation of a particular boundary discretization.

1.2.4 Model Hierarchy and Comparison Boundary

Reduced hemodynamic models form a hierarchy of modeling roles rather than a simple accuracy ranking. A higher-dimensional model becomes a comparison only after the kinematic and continuum assumption package, geometry, wall response, rheology, inlet and outlet data, numerical resolution, output operators, and discrepancy metrics are matched.

The table should be read together with Definition 1.26. Spatial dimension and wall coupling are separate choices: a 3D computation can be fixed-wall or resolved FSI, and a 1D computation can be rigid-tube, elastic-wall, or coupled to upstream/downstream models. Consequently, an FSI comparison claim requires

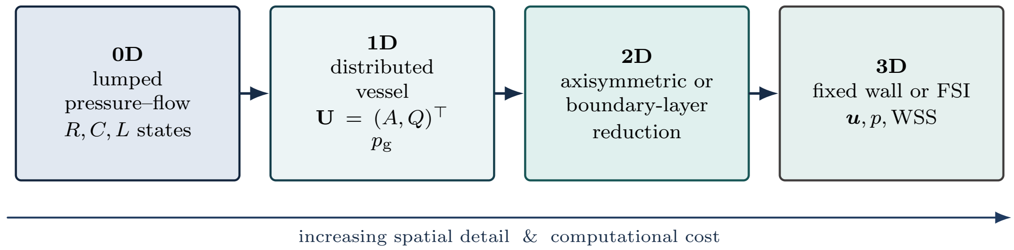

Figure 5: Model hierarchy for reading comparison claims. The present manuscript selects a 1D reduced forward model; 3D fixed-wall, resolved FSI, clinical, and independent-solver comparisons require separately matched evidence.

Table 1: Hemodynamic model taxonomy for the idealized stenosis report.

Model family	**Typical role**	**Common states or outputs**	**Report boundary**
0D lumped circulation	Outlet loads, Windkessel context, and network hemodynamic summaries	Compartment pressures, flows, resistances, compliances, inertances	Useful for boundary conditions, not for local stenosis geometry or resolved wall traction
Classical 1D compliant-vessel models	Distributed area-flow description after cross-sectional averaging	$A(z,t)$, $Q(z,t)$, $p_g(z,t)$, wave speed, pressure drop	Main reduced model tier under declared continuum, wall, friction, inlet, and outlet assumptions
Stenosis-aware 1D and pressure-loss ROMs	Idealized stenosis trends, variable-radius corrections, or algebraic loss estimates	Stenosis profile, severity, flow rate, Δp, pressure ratio, reduced shear proxy	Reduced-order trend context only unless matched against resolved or clinical evidence
2D/3D fixed-wall and resolved-FSI models	Higher-fidelity local flow and wall-motion comparison	Velocity, pressure, wall traction, WSS, wall displacement, mesh and time diagnostics	Reference-tier context unless the same continuum setting, geometry, rheology, wall, boundary data, and outputs are compared directly
Numerical reference problems for 1D solvers	Scheme comparison, reproducibility expectations, and implementation context	Grid, timestep, flux and source choices, boundary states, field and scalar discrepancies	Evidence standard; does not grant solver parity to the present code without a matched comparison

the wall descriptor $\mathcal{W}$, rheology descriptor $\mathcal{R}$, boundary data, interface conditions, and output operators to be matched; it does not follow from the flux/source notation or from model dimension alone.

Across these categories, the recurring descriptors are the kinematic/continuum setting, stenosis geometry, wall stiffness or compliance, density, viscosity or effective-viscosity law, inlet waveform, outlet pressure or load, pressure-reference convention, and tap locations. The recurring outputs are pressure drop, reduced pressure ratio, area, flow, radius, and either a reduced wall-friction/shear proxy or a resolved traction-based WSS quantity [1, 9, 10] [12, 13]. The report therefore treats the selected 1D area-flow model as the controlled forward map for the first numerical study, while reserving 3D/FSI agreement, clinical calibration, and independent-solver parity for later evidence. The pressure–flow motivation is stated after the continuum and model-hierarchy setup so that pressure-drop, pressure-ratio, and shear quantities are introduced as model outputs with declared conventions, not as clinical measurements or validation targets.

1.3 Clinical Pressure–Flow Motivation

Anatomy–function distinction. Coronary stenosis is the motivating lesion class because a visible narrowing does not automatically imply a flow-limiting obstruction. Clinical assessment therefore distinguishes anatomical stenosis detection from functional pressure–flow assessment [16, p. 4401] [17, Ch. 1, p. 15, Fig. 1.14, Ch. 2, Secs. 2.1–2.2, pp. 22–23, Ch. 4, Secs. 4.1 and 4.3, pp. 56–57] [18, Sec. 1, Introduction, pp. 390–391]. Coronary lesion-imaging studies likewise show that anatomy-side lesion description carries local geometry and length context, but those descriptors still remain distinct from the functional pressure–flow outputs treated here [19, Abstract, Methods, Table 2, Tables 3–5]. This anatomy–function split motivates reporting pressure drop, reduced pressure ratio, and reduced shear information rather than treating stenosis severity $s_{\max}$ as the output. Here pressure-drop, pressure-ratio, and shear-proxy quantities are reduced-output conventions for the model framework. They are not all demonstrated outputs in Section 2, whose present comparison reports velocity diagnostics only. Figure 6 summarizes this motivation as a schematic distinction between visible narrowing and pressure–flow loss.

Clinical fractional flow reserve (FFR) is an invasive pressure-wire index, conventionally interpreted in its measurement setting as distal coronary pressure divided by aortic pressure during maximal hyperemia [17, Ch. 5, Sec. 5.5, pp. 72–73] [20, Abstract, Background]. FAME lesion-level evidence documented the discordance between angiographic and functional severity and is the clinical source for the familiar FFR ≤ 0.80 decision context. [21, Abstract, Results and Conclusions]. Computed tomography-derived FFR (CT-FFR) sources supply noninvasive comparison vocabulary within their reported study designs: they compare pressure–flow quantities inferred from coronary computed tomography angiography (CCTA)-derived anatomy against invasive FFR or related reference measurements [22, Abstract, Methods, Results and Conclusions] [18, Abstract, Sec. 1, Introduction, Secs. 1.1–1.2, pp. 390–392].

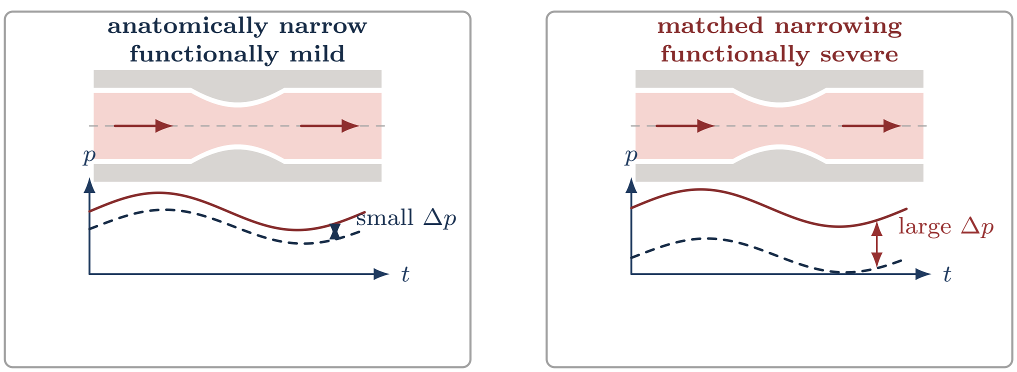

Figure 6: Pressure–flow motivation for the reduced-output contract. The schematic distinguishes an anatomical narrowing from the pressure-loss signature used by reduced-model outputs.

For the selected model gauge-pressure field p_g, define the observed proximal and distal pressure traces by

$$P_{\text{prox}}(t) \;:=\; \mathcal{P}_{\text{prox}}[p_g(\cdot,t)], \qquad P_{\text{dist}}(t) \;:=\; \mathcal{P}_{\text{dist}}[p_g(\cdot,t)].$$

The pressure-drop waveform is the additive-datum-invariant quantity

$$\Delta p(t) \;:=\; P_{\text{prox}}(t) - P_{\text{dist}}(t).$$

Unlike $\Delta p(t)$, a pressure ratio is not invariant under additive pressure shifts, so the ratio convention must declare a common additive pressure reference. The manuscript's convention adds that reference after the pressure observations have been taken. With the declared p_{ref}, define

$$P^{\mathrm{ratio}}_{\mathrm{prox}}(t) \;:=\; p_{\mathrm{ref}} + P_{\mathrm{prox}}(t), \qquad P^{\mathrm{ratio}}_{\mathrm{dist}}(t) \;:=\; p_{\mathrm{ref}} + P_{\mathrm{dist}}(t).$$

If a selected observation functional preserves constant shifts, then these ratio-pressure traces agree with observing $p_{\mathrm{ref}} + p_{\mathrm{g}}(\cdot, t)$ directly. The definition above does not require that property; it fixes the ratio convention at the level of observed scalar traces. For an admissible ratio-output specification whose observed proximal and distal ratio-pressure traces are integrable on the window and whose declared denominator rule makes the proximal averaged denominator well defined and nonzero, the reduced pressure ratio is

$$\mathrm{PR}_{1D}(\pi_{\mathrm{PR}}) \;:=\; \frac{\langle P^{\mathrm{ratio}}_{\mathrm{dist}}(t)\rangle_{\mathcal{T}_{\mathrm{HF}}}}{\langle P^{\mathrm{ratio}}_{\mathrm{prox}}(t)\rangle_{\mathcal{T}_{\mathrm{HF}}}}.$$

The 1D specialization used later is the case in which $\mathcal{P}_{\mathrm{prox}}$ and $\mathcal{P}_{\mathrm{dist}}$ are evaluated from the declared axial pressure taps or cross-sectional pressure averages; see Figure 7.

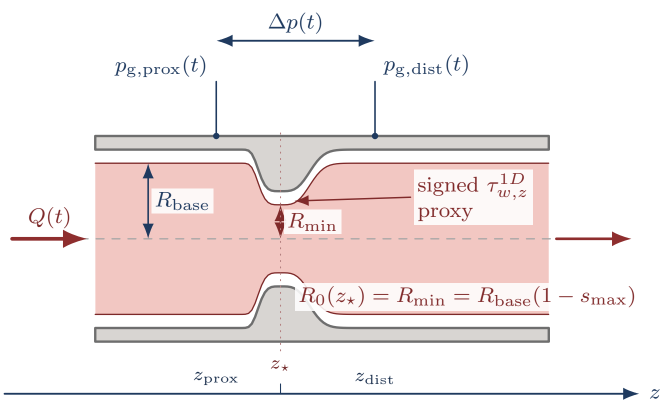

Figure 7: Schematic tap and output convention for an idealized stenosis, not drawn from a computed case. The taps define $\Delta p(t)$; π_{PR} additionally declares the averaging window, pressure reference, and denominator rule needed for $\mathrm{PR}_{1D}(\pi_{\mathrm{PR}})$.

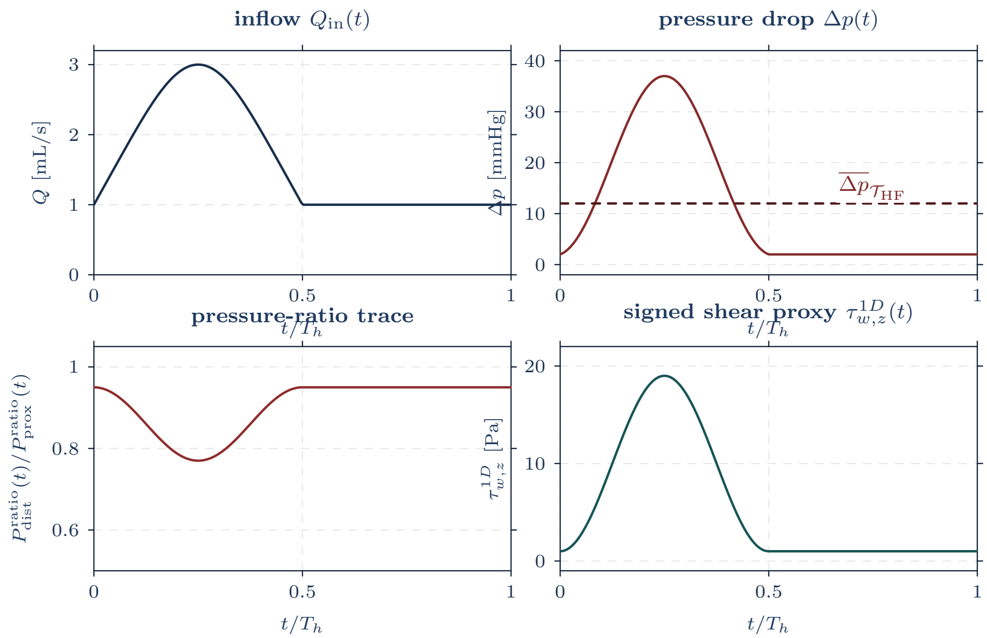

Figure 8: Reduced-output signals over one schematic cardiac cycle. The imposed inflow trace, pressure-drop waveform, metadata-dependent pressure ratio, and signed 1D shear proxy illustrate the declared output semantics, not simulated data or acceptance evidence.

Resolved wall shear stress (WSS) and reduced shear proxy. Pressure ratios and pressure drops summarize lesion-scale functional loss in the selected model tier. Wall shear stress describes local wall loading in a resolved state. These roles are related but not interchangeable. When the stress trace is well defined, resolved WSS is the tangential part of the wall traction obtained from a resolved stress tensor and wall normal; a signed component additionally requires a chosen wall-tangent direction [23, Abstract, Sec. 2, pp. 8–10]. The reduced shear proxy used later is therefore a tier-specific closure diagnostic, separate from resolved traction-based WSS.

2 Preliminary Resolved-Velocity Diagnostic at $T = 1$

This section records the numerical comparison produced by the Julia implementation of the stenosis-aware 1D area–flow model against the available resolved 3D velocity data. The comparison is an output-operator check, not a clinical validation claim and not an exact cross-section quadrature of the 3D tetrahedral solution. The 3D diagnostic averages node-centered axial velocity over thin axial slabs and compares those values with the interpolated 1D recorded mean velocity Q/a, where $a = R^2$ is the solver coordinate. Radial profiles use the same slab nodes binned by $\sqrt{x^2+y^2}/R_0(z)$ and compare them with the 1D closure profile. The tables and figures in this section are baseline default Newtonian records; non-Newtonian rheology descriptors require separately generated comparison records.

> **Section 2 notation warning.** In this section only, the implementation variable named `A` is the solver coordinate $a = R^2$, not physical cross-sectional area. The physical area is $A_{\text{phys}} = \pi a$, and the reported mean velocity is Q/a, equivalently $Q_{\text{phys}}/A_{\text{phys}}$ under the recorded π-scaling.

All numbers below come from the local record directory `simulations/output/3d_comparison/full_t1_native_nx400/`, interpreted under Convention G.1. The record is generated by the available resolved-3D comparison entry point in `resolved3d.jl`, with overwrite enabled and the output directory set to that record path. The comparison uses default parameters with final time $T = 1.0$ s and the geometry-rest initial condition, with $N = 400$ finite-volume cells on $L = 6$ cm ($\Delta z = 0.015$ cm), the native SSPRK3 backend, MUSCL finite volume with the minmod limiter, timestep cap 10^{-5} s, and CFL cap 0.45. The initial state is geometry-rest, $a = R_0^2$ and $Q = 0$. The inlet boundary uses the parabolic-profile mean velocity $\bar{u}_{\text{in}} = 0.5(45) = 22.5$ cm/s and imposes $Q_{\text{in}} = R_0(0)^2\bar{u}_{\text{in}}$; the outlet fixes $a_{\text{out}} = R_0(L)^2$ and computes the ghost flow from the outgoing Riemann invariant. The two 3D source identifiers are the default upstream XDMF paths `simulations/data/3d/canic_case3/77/velocity.xdmf` and `simulations/data/3d/canic_case3/60/velocity.xdmf`, with severities 23% and 40%, target time 1.0 s, and tolerance 10^{-3} s. The recorded XDMF time for both included cases is 0.9994999999999453 s. Section means use 200 equally spaced axial targets and slab half-width $\max(\Delta z_{\text{sec}}/2, 0.015) = 0.0150753769$ cm; radial profiles use the default slices $z = 1.951, 2.451, 2.951$ cm and 20 radial bins in $\sqrt{x^2+y^2}/R_0(z)$. The raw XDMF/HDF5 inputs are ignored by git and are not present in this checkout; the local audit trail available here is the record directory above plus the copied static plot data in `figures/static/static/data/stenosis-comparison/`.

Table 2: Availability and run-diagnostic status for the local $T = 1$ comparison record. The controls are known from the recorded setup; per-step diagnostics are not persisted in the available ignored CSV/SVG record.

Case	Severity	$\Delta t_{\max}$ (s)	CFL cap	3D time (s)	Missing per-step diagnostics
77	23%	10^{-5}	0.45	0.9994999999999453	steps, realized max CFL, min a, residual summary, positivity failures
60	40%	10^{-5}	0.45	0.9994999999999453	steps, realized max CFL, min a, residual summary, positivity failures

In the active solver, the code variable A is the radius-squared coordinate $a = R^2$, not the physical area $A_{\text{phys}} = \pi a$. The recorded flow variable is therefore the corresponding π-scaled flow: $Q_{\text{phys}} = \pi Q$. All comparison velocities in this section use the recorded field `velocity(result)` $= Q/a$, which equals $Q_{\text{phys}}/A_{\text{phys}}$ under this scaling. The plots and tables should not be read as using the stored Q divided by A_{phys}.

The active Section 2 closure is the Canic extended radius-squared finite-volume record with $\mathcal{W}_{\text{elastic1D}}$, the elastic pressure–area wall law in Definition 1.29, `newtonian` rheology ($\nu = 0.04$ cm^2/s), and the parabolic velocity profile $\mathcal{P} =$ `parabolic`, so $\alpha_{\mathcal{P}} = 4/3$ and $g_{\mathcal{P}} = 4$. The Canic variable-radius corrections are active in this record: the flux uses $\alpha_{\mathcal{P}} + \alpha_c(R_0')$, and the source includes the p_2, R_0', R_0'', and α_c' terms described in Appendix G.

The two included upstream states are case 77, treated as a 23% stenosis, and case 60, treated as a 40% stenosis. Both XDMF files report $t = 0.9994999999999453$, within 5.0×10^{-4} of $T = 1$. Upstream case 50 was not used because its XDMF time is $t = 1.4994999999998904$, outside the 10^{-3} tolerance for the $T = 1$ comparison.

Table 3: Section-mean and radial-profile discrepancy summary for the $T = 1$ resolved-node comparison. Velocity errors are in cm/s.

Case	Severity	Sections	Min nodes	Mean $\lvert e_s \rvert$	Max $\lvert e_s \rvert$	Mean rel.	Max rel.	Mean $\lvert e_r \rvert$	Max $\lvert e_r \rvert$
77	23%	200	64	7.471	26.26	0.4534	1.622	10.03	23.01
60	40%	200	43	9.440	47.48	0.5141	2.458	15.35	41.72

Figure 9 shows the section-mean comparison. The 1D model captures the smooth upstream and downstream trend but overpredicts the highest-velocity region near the stenosis, especially for the 40% case. The largest section errors occur for case 60 around $z = 2.4$–2.8 cm, where the 1D mean velocity is between 58 and 67 cm/s while the node-slab 3D mean is between 19 and 27 cm/s. Because the largest relative errors exceed unity near the stenosis, this record is not acceptable as validation of the 1D model in the throat region. It is still useful as a diagnostic localization: the mismatch concentrates where the reduced velocity closure, idealized 1D boundary state, and simple node-slab output operator are least matched to the resolved 3D field.

Table 4: Five largest section-mean absolute errors across the two included $T = 1$ cases. Velocities and errors are in cm/s.

Rank	Case	z (cm)	u_{1D}	u_{3D}	$\lvert e \rvert$	Rel.	Nodes
1	60	2.714	66.79	19.31	47.48	2.458	72
2	60	2.774	63.74	24.57	39.17	1.594	60
3	60	2.412	61.90	23.51	38.39	1.633	67
4	60	2.563	63.71	27.15	36.56	1.347	72
5	60	2.834	57.59	21.68	35.92	1.657	82

The radial-profile diagnostics in Figure 10 show the same pattern at the profile level. For both cases, the throat slice $z = 2.451$ cm has fewer populated radial bins because fewer 3D nodes lie in the narrowest

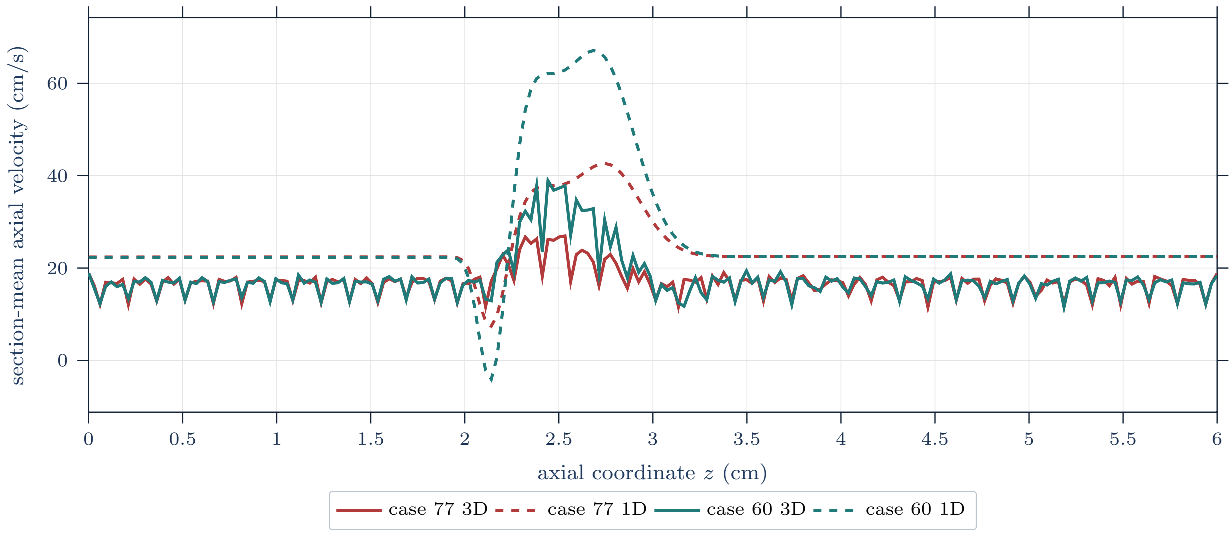

Figure 9: Section-mean axial velocity comparison at $T = 1$. Solid curves are resolved 3D node-slab means; dashed curves are interpolated 1D Q/a, equivalently $Q_{\text{phys}}/A_{\text{phys}}$ under the recorded π-scaling.

cross-section slab. The largest radial-profile discrepancy is downstream of the throat at $z = 2.951$ cm, where case 60 reaches a maximum per-bin absolute error of 41.72 cm/s. The dashed 1D curves use the recorded velocity-profile closure evaluated from the local section mean $\bar{u} = Q/a$. For the power-family diagnostic,

$$u_{\mathcal{P}}(r) = \frac{\gamma+2}{\gamma}\,\bar{u}\left(1 - \left(\frac{r}{R}\right)^{\gamma}\right), \qquad \alpha_{\mathcal{P}} = \frac{\gamma+2}{\gamma+1}.$$

The former $\alpha = 1.1$ record is therefore the $\gamma = 9$ member of this family; the parabolic profile is $\gamma = 2$ and $\alpha_{\mathcal{P}} = 4/3$. Flat-profile diagnostics use $u_{\mathcal{P}}(r) = \bar{u}$ with $\alpha_{\mathcal{P}} = 1$.

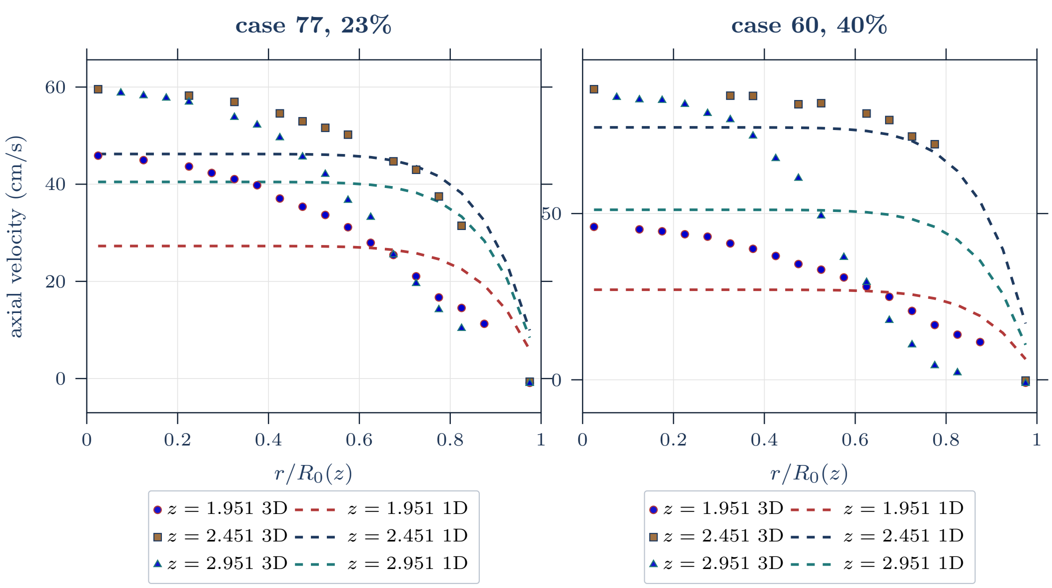

Figure 10: Radial velocity profile comparison at the default throat-centered slices. Markers are resolved 3D radial-bin means; dashed curves are the 1D closure profile reconstructed from the local Q/a.

Table 5: Radial-profile discrepancy by case and slice. Errors are in cm/s.

Case	**z (cm)**	**Populated bins**	**Mean $\|e_r\|$**	**Max $\|e_r\|$**
77	1.951	17	9.404	18.60
77	2.451	12	6.949	13.33
77	2.951	16	13.01	23.01
60	1.951	18	10.04	18.90
60	2.451	10	7.562	17.17
60	2.951	17	25.55	41.72

These discrepancies should be read in light of the comparison boundary in Section 1.2.4. The present 1D solver is compared to a resolved 3D velocity field through simple node-slab output operators. The result is useful for locating where the reduced velocity closure departs from the resolved state, but it does not by itself identify whether the discrepancy comes from the 1D closure, the boundary data, the 3D post-processing operator, or differences between the numerical formulations. In summary, the $T = 1$ comparison shows that the recorded 1D solution gives a controlled reduced-output diagnostic whose largest deviations are localized near and downstream of the stenosis throat, especially for the 40% case. It does not establish resolved-3D accuracy, clinical validity, or independence from the chosen 3D node-slab output operator and boundary realization. The next defensible study step is therefore a matched comparison record that restores the raw 3D inputs, records the source hashes or upstream release identifiers, and reruns the same output operators while varying only one declared closure or boundary choice at a time.

3 Conclusions and Limitations

This report established a mathematical vocabulary for idealized stenotic blood flow and used it to state a bounded numerical comparison. The continuum portion connected the flow map, material derivative, transport theorem, balance laws, stress closure, Navier–Stokes equations, and one-dimensional area–flow reduction. The model-hierarchy discussion then separated reduced pressure ratios, reduced shear proxies, and velocity diagnostics from clinical measurements or resolved three-dimensional wall quantities.

The numerical comparison demonstrated that the selected one-dimensional solver can be placed in a declared diagnostic relationship with available resolved 3D velocity data through section-mean and radial-profile output operators. For the two included idealized cases at $T = 1$, the 1D solution gives broadly similar upstream and downstream velocity trends but overpredicts the near-throat velocity, especially for the 40% case. The largest errors are therefore not a minor plotting discrepancy; they occur in the region where the reduced velocity closure, boundary realization, and simple node-slab 3D post-processing operator are least interchangeable.

The comparison does not validate the 1D model against resolved 3D flow, does not establish clinical validity, and does not demonstrate pressure-drop or pressure-ratio accuracy. It is a preliminary velocity diagnostic for a local record whose raw XDMF/HDF5 inputs are not present in the checkout. The pressure-ratio and shear-proxy conventions are included to fix the output vocabulary for later reduced-order studies, not as completed numerical evidence in this report.

A stronger follow-up comparison would restore or archive the raw 3D inputs, record source hashes or release identifiers, preserve per-run diagnostics such as time-step counts, CFL extrema, positivity checks, and residual summaries, and then vary one declared closure or boundary choice at a time. That study would also need pressure-drop and pressure-ratio outputs if the clinical motivation is to be tested rather than only framed.

A Acronyms and Abbreviations

General abbreviations.

a.e.	almost everywhere	bpm	beats per minute
e.g.	*exempli gratia* (for example)	i.e.	*id est* (that is)

Clinical, modeling, and numerical acronyms.

CAS	coronary artery stenosis	FSI	fluid–structure interaction
CCTA	coronary computed tomography angiography	RBC	red blood cell
CT-FFR	computed tomography-derived fractional flow reserve	FFR	fractional flow reserve
IVUS	intravascular ultrasound	WSS	wall shear stress
OCT	optical coherence tomography	CFD	computational fluid dynamics
ODE	ordinary differential equation	PDE	partial differential equation
IC	initial condition	BC	boundary condition
FDM	finite difference method	FEM	finite element method
FVM	finite volume method	ROM	reduced-order model
0D	zero-dimensional	1D	one-dimensional
2D	two-dimensional	3D	three-dimensional

B Mathematical Notation

Notation B.1 (Report mathematical notation)**.** This appendix collects the mathematical notation used throughout this report.

Symbol	**Meaning**
$:=$	defines
$\equiv$	notational equivalence / identified with
$\mathbb{R}$	real numbers
$\mathbb{R}^+$	strictly positive real numbers
$\mathbb{R}^-$	strictly negative real numbers
$\mathbb{R}^n$	n-dimensional real vector space
$\mathbb{N}$	natural numbers
$\mathbb{N}_0$	nonnegative integers $\{0, 1, 2, \ldots\}$
$\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_n}\}$	general basis of $\mathbb{R}^n$
$\{\boldsymbol{e_1}, \ldots, \boldsymbol{e_n}\}$	standard basis of $\mathbb{R}^n$
$[n]$	index set $\{1, 2, \ldots, n\} \subset \mathbb{N}$ for $n \in \mathbb{N}$
I_T	open time interval $(0, T_{\text{final}})$ for transient interior-time arguments
$\overline{I}_T$	closed time interval $[0, T_{\text{final}}]$ for initial-time data and flow maps
K	compact space-time rectangle, such as $[0, \ell] \times [0, T_{\text{final}}]$, used for output-field norms when locally stated
$\Omega \subset \mathbb{R}^n$	bounded domain
$\overline{\Omega}$	the closure of Ω
$\partial\Omega$	the boundary of Ω
$\Omega_B(t)$	spatial blood domain at time t
$\mathcal{Q}_T$	space-time fluid region $\{(t, \boldsymbol{x}) : t \in I_T,\ \boldsymbol{x} \in \Omega_B(t)\}$
$C^k(\Omega)$	functions with continuous classical derivatives through order k in Ω
$C^k(\overline{\Omega})$	functions whose derivatives through order k extend continuously to $\overline{\Omega}$
$C^k_c(\Omega)$	k times continuously differentiable functions with compact support in Ω
$C^\infty_c(\Omega)$	smooth functions with compact support in Ω

Symbol	Meaning
$C(K)$	continuous functions on a stated compact set K
$L^p(\Omega)$	Lebesgue space of p-integrable functions modulo equality a.e.
$d\boldsymbol{x}$	Lebesgue volume element on $\mathbb{R}^n$
$dS_{\boldsymbol{x}}$	surface measure element on $\partial\Omega \subset \mathbb{R}^n$
dV	3D volume element on domain $\Omega \subset \mathbb{R}^3$
dS	3D surface element on boundary $\partial\Omega \subset \mathbb{R}^3$
∇	gradient operator
$\partial_{x_i}\phi$	partial derivative of ϕ with respect to coordinate x_i
$\partial_t\phi$, $\partial_z f$, $\partial_A p$	compact partial derivatives with respect to named coordinates t, z, and A
$\partial_r u$, $\partial_\theta u$	compact partial derivatives in cylindrical coordinates
$\Delta\phi = \nabla \cdot \nabla\phi$	Laplacian of scalar field ϕ
div	divergence of a vector field
div	row-wise divergence of a tensor field
$\boldsymbol{v}_i$	i-th component of vector $\boldsymbol{v}$
$\boldsymbol{S} : \boldsymbol{T}$	Frobenius contraction $\sum_{i,j} S_{ij}T_{ij}$ for second-order tensors
$\langle \cdot,\ \cdot \rangle_X$	inner product on vector space X
$\langle \boldsymbol{u},\ \boldsymbol{v} \rangle \ \equiv\ \langle \boldsymbol{u},\ \boldsymbol{v} \rangle_{\mathbb{R}^n}$	inner product of vectors $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$
$\partial_{\hat{\boldsymbol{n}}}\phi$	normal derivative $\langle \nabla\phi,\ \hat{\boldsymbol{n}} \rangle$ of scalar field ϕ on $\partial\Omega$
$\lVert\cdot\rVert_X$	norm on a specified normed space X; bare $\lVert\cdot\rVert$ is local shorthand
$\lVert\cdot\rVert_{\mathrm{F}}$	Frobenius norm of a matrix or second-order tensor

C Domain Specific Notation

Definition C.1 (SI unit convention)**.** Unless otherwise stated, physical units are SI: length in meters, time in seconds, density in $\mathrm{kg/m^3}$, pressure in Pa, dynamic viscosity in $\mathrm{Pa\cdot s}$, and kinematic viscosity in $\mathrm{m^2/s}$.

Convention C.2 (Centimeter–second numerical-record convention)**.** The SI convention above governs continuum notation and unqualified analytical parameters. The active Julia solver records and the Section 2 tables instead use the centimeter–gram–second numerical convention inherited from the implementation: lengths are in cm, time in s, density in $\mathrm{g/cm^3}$, velocity in cm/s, pressure and stress in $\mathrm{dyn/cm^2}$, dynamic viscosity in $\mathrm{g/(cm\,s)} = \mathrm{dyn\,s/cm^2}$, and kinematic viscosity in $\mathrm{cm^2/s}$. Thus the default solver geometry parameters L, $R_{\max}$, z_a, z_s, σ_a, and κ are centimeter quantities, while the stenosis localization λ in $\exp(-\lambda g^4)$ has units $\mathrm{cm^{-4}}$. The active solver state coordinate $a = R^2$ has units $\mathrm{cm^2}$; when a physical circular area is meant it is $A_{\text{phys}} = \pi a$, also reported in square centimeters.

Output fields carry their display units in their names. In particular, `z_cm` and `R0_cm` are geometry fields, `A_cm2` is the recorded area-coordinate field in $\mathrm{cm^2}$, and `Q_cm3_s` is the flow-coordinate field paired with a in nominal $\mathrm{cm^3/s}$. For the active radius-squared solver convention, the corresponding physical circular-area flow is $Q_{\text{phys}} = \pi Q$, while `uavg_cm_s` records $Q/a = Q_{\text{phys}}/A_{\text{phys}}$. The Section 2 velocity/error columns are in cm/s, `pressure_dyn_cm2` and initial-condition pressure-drop fields are in $\mathrm{dyn/cm^2}$, and `nu_eff_cm2_s` is in $\mathrm{cm^2/s}$. A pressure supplied through a Pa-valued command-line option is converted before storage using $1\,\mathrm{Pa} = 10\,\mathrm{dyn/cm^2}$. Any value from a numerical record used in an SI formula must first be converted; for example $1\,\mathrm{cm} = 10^{-2}\,\mathrm{m}$, $1\,\mathrm{cm^2/s} = 10^{-4}\,\mathrm{m^2/s}$, $1\,\mathrm{g/(cm\,s)} = 10^{-1}\,\mathrm{Pa\cdot s}$, and $1\,\mathrm{dyn/cm^2} = 10^{-1}\,\mathrm{Pa}$.

Convention C.3 (Pressure-reference and ratio convention)**.** Pressure-ratio outputs are reported only with a recorded pressure reference p_{ref} and a gauge pressure p_{g} in the same reference frame. The reference-consistent pressure used in ratios is

$$p_{\text{ratio}} \;:=\; p_{\text{ref}} + p_{\text{g}}.$$

The ratio-output specification is

$$\pi_{\text{PR}} \;:=\; (z_{\text{prox}}, z_{\text{dist}}, \mathcal{T}_{\text{HF}}, p_{\text{ref}}).$$

The subscript HF labels a declared high-flow averaging window in the study record; it is not a clinical hyperemia model unless an external measurement comparison is separately specified [17, Ch. 5, Sec. 5.5, pp. 72–73] [20, p. 1703]. The ratio-output specification is admissible for a computed pressure trace when $\mathcal{T}_{\text{HF}} \subset I_T$ is Lebesgue-measurable with $0 < |\mathcal{T}_{\text{HF}}| < \infty$, the proximal and distal p_{ratio} traces are integrable on that window, and

$$\langle p_{\mathrm{ratio}}(z_{\mathrm{prox}}, t)\rangle_{\mathcal{T}_{\mathrm{HF}}} \neq 0.$$

For an admissible π_{PR}, the corresponding nonclinical pressure ratio is

$$\mathrm{PR}_{1D}(\pi_{\mathrm{PR}}) \;:=\; \frac{\langle p_{\mathrm{ratio}}(z_{\mathrm{dist}}, t)\rangle_{\mathcal{T}_{\mathrm{HF}}}}{\langle p_{\mathrm{ratio}}(z_{\mathrm{prox}}, t)\rangle_{\mathcal{T}_{\mathrm{HF}}}}.$$

For an integrable scalar trace,

$$\langle f\rangle_{\mathcal{T}_{\mathrm{HF}}} \;:=\; \frac{1}{|\mathcal{T}_{\mathrm{HF}}|}\int_{\mathcal{T}_{\mathrm{HF}}} f(t)\,dt.$$

This bracket denotes time averaging, not the inner product notation in Appendix B. An FFR-like shorthand refers to this reduced ratio only for records carrying an admissible π_{PR}; the shorthand does not add a clinical measurement model [17, Ch. 5, Sec. 5.5, pp. 72–73] [20, p. 1703]. If π_{PR} is absent or inadmissible, the pressure ratio is outside the reported output space.

Symbol	Meaning	Units
R	vessel radius; vessel diameter is $d_v := 2R$	m
d_v	vessel diameter	m
η	dynamic viscosity	$\mathrm{Pa} \cdot \mathrm{s}$
$\nu := \eta/\rho$	kinematic viscosity	m^2/s
$\eta_{\mathrm{eff}}(\dot{\gamma})$	effective dynamic viscosity for generalized Newtonian models	$\mathrm{Pa} \cdot \mathrm{s}$
$\nu_{\mathrm{eff}}(\dot{\gamma}) := \eta_{\mathrm{eff}}(\dot{\gamma})/\rho$	effective kinematic viscosity after density scaling	m^2/s
τ	shear stress	Pa
$\dot{\gamma}$	shear rate	s^{-1}
ρ	density field	kg/m^3
p	pressure field	Pa
$\boldsymbol{u}$	velocity field	m/s
$D(\boldsymbol{u})$	rate-of-deformation tensor, $D(\boldsymbol{u}) := \frac{1}{2}\left(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^{\top}\right)$	s^{-1}
$W_0 := d_v\sqrt{\frac{\omega\rho}{\eta}}$	Womersley number, diameter-based convention	$-$
Re	Reynolds number	$-$
Pe	Péclet number	$-$
c_{conc}	concentration field when a transport scalar is discussed	$-$
D_m	molecular diffusion coefficient	m^2/s
t	time	s
T or T_{final}	terminal time; use T_{final} where it could be confused with stress tensors	s
ω	angular frequency	rad/s
(r, θ, z)	cylindrical coordinates used only when 3D coordinate forms are written	$\mathrm{m}, -, \mathrm{m}$
$(\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_z)$	cylindrical orthonormal basis vectors	$-$
τ_{ij}	cylindrical viscous-stress component	Pa
$\boldsymbol{f_b}$	body force per unit mass	m/s^2
$\rho\boldsymbol{f_b}$	body force per unit volume	N/m^3

Reduced-model and forward-map notation.

Symbol	Meaning	Units
$A(z,t)$	cross-sectional lumen area in the 1D model	m^2
$Q(z,t)$	volumetric flow rate through a cross-section	m^3/s
$\bar{u}(z,t) := Q/A$	section-mean axial velocity	m/s
$I_z = (0, L)$	axial single-vessel interval for a 1D formulation	m
$\mathbf{U} = (A, Q)^\top$	conservative area-flow state vector	$(\mathrm{m}^2, \mathrm{m}^3/\mathrm{s})$
$\mathcal{P}$	cross-sectional velocity-profile descriptor used by the 1D closure; admitted implementation values include flat, parabolic, and power-family profiles	mixed
$\alpha_{\mathcal{P}}$	momentum-flux correction induced by the recorded profile $\mathcal{P}$; the parabolic default uses $\alpha_{\mathcal{P}} = 4/3$	–
γ	power-profile exponent in $u(r) = ((\gamma+2)/\gamma)\bar{u}(1-(r/R)^\gamma)$	–
$g_{\mathcal{P}}$	profile shear-rate factor used by the 1D rheology proxy; for power profiles $g_{\mathcal{P}} = \gamma + 2$, while flat profiles record this value explicitly	–
$R(A(z,t)) = \sqrt{A(z,t)/\pi}$	current reduced lumen radius derived from the 1D area	m
$R_0(z)$	reference stenosed radius used by the wall law	m
$R_{\mathrm{base}}(z)$	nominal healthy baseline radius before applying $s(z)$	m
$\mathcal{R}$	rheology descriptor in a_{1D}; distinct from radius symbols R, R_0, and R_{base}; admitted implementation values are `newtonian`, `carreau`, `carreau-yasuda`, `casson`, and `power-law`	mixed
$\mathcal{W}$	wall or FSI closure descriptor; the selected 1D report model uses $\mathcal{W}_{\mathrm{elastic1D}}$, while fixed-wall, multiscale-coupled, and resolved-FSI models require separate declarations	mixed
$s(z)$	dimensionless stenosis profile, with $0 \le s(z) < 1$	–
$A_0(z) = \pi R_0(z)^2$	wall-law reference area after applying the stenosis profile	m^2
$s_{\max}$	maximum fractional radius-reduction parameter in $s(z)$	–
$g(z)$	smooth asymmetric stenosis-kernel coordinate	m
$\psi(z) = \exp(-\lambda g(z)^4)$	smooth rapidly localized stenosis kernel used by the baseline profile $s(z) = s_{\max}\psi(z)$	–
z_a	center of the Gaussian asymmetry term in $g(z)$	m
z_s	axial shift parameter in $g(z)$	m

Symbol	Meaning	Units
σ_a	width parameter for the Gaussian asymmetry term in $g(z)$	m
κ	amplitude of the Gaussian asymmetry term in $g(z)$	m
λ	localization strength in $\psi(z) = \exp(-\lambda g(z)^4)$	m^{-4}
$\beta(z)$	wall-stiffness parameter in $p_\mathrm{g} = p_\mathrm{ext} + \beta A_0^{-1}(\sqrt{A} - \sqrt{A_0})$	$\mathrm{Pa} \cdot \mathrm{m}$
$p_\mathrm{ext}(z,t)$	external-pressure datum in the wall-law gauge convention; baseline value $p_\mathrm{ext} \equiv 0$ unless declared otherwise	Pa
$\Psi(A,z,t)$ or $\Psi_\mathcal{W}(A,z,t)$	elastic flux potential satisfying $\partial_A \Psi_\mathcal{W} = (A/\rho)\partial_A p_\mathrm{g}^\mathcal{W}$ and normalized by $\Psi_\mathcal{W}(A_0(z),z,t) = 0$ when that normalization is declared	$\mathrm{m}^4/\mathrm{s}^2$
$\mathbf{F}$	area-flow flux map; not a residual or objective functional	problem-dependent
$\mathbf{S}$	source-term map in the 1D balance-law template	problem-dependent
$S_\mathrm{geom}^\mathcal{W}$	fixed-area wall/geometry source associated with the selected reduced wall closure $\mathcal{W}$	$\mathrm{m}^3/\mathrm{s}^2$
$S_f^\mathcal{R}$	wall-friction source associated with the selected rheology or effective-viscosity closure $\mathcal{R}$	$\mathrm{m}^3/\mathrm{s}^2$
$\widehat{\mathbf{F}}_{i+1/2}^n$	numerical flux at a finite-volume cell interface and time level	problem-dependent
$\widehat{\mathbf{S}}_i^n$	cell source quadrature used by a computed finite-volume scheme	problem-dependent
$\Delta z_i, \Delta t^n$	finite-volume cell length and timestep in a computed solve	m, s
$\dot{\gamma}_{1D}$	implementation shear-rate proxy $g_\mathcal{P}\|Q\|/(a\sqrt{a})$, where $a = R^2$	s^{-1}
$\nu_\mathrm{eff}^\mathcal{R}$	closure-indexed effective kinematic viscosity used in the current reduced viscous/source and pressure-correction terms	m^2/s
C_f or $C_f(z,A)$	wall-friction coefficient in the 1D source $-C_f Q/A$; constant-viscosity value $C_f = 8\pi\eta/\rho$	m^2/s
$\eta_\mathrm{fric}(z,t)$	dynamic viscosity used by the reduced wall-friction and signed-shear-proxy formula; $\eta_\mathrm{fric} = \eta$ for constant-viscosity records and $\eta_\mathrm{fric} = \eta_\mathrm{eff}^\mathcal{R}(\dot{\gamma}_{1D})$ for admitted non-Newtonian descriptor records	$\mathrm{Pa} \cdot \mathrm{s}$
c_wave or c	1D pulse-wave speed, $c^2 = (A/\rho)\partial_A p_\mathrm{g}$ when no concentration field is in scope	m/s

Symbol	Meaning	Units
$e \in \mathcal{E}$	expert/closure record for a closure-indexed 1D forward-map family; suppress e only for single-expert baseline records	mixed
$\mathcal{G}_{1D,h}^{r_h,e}$	computed fixed-grid 1D map indexed by numerical record r_h and expert record e	mixed

Symbol	**Meaning**	**Units**
p_{3D}	continuum pressure in the 3D/NS setting	Pa
$p_{3D,\mathrm{g}}$	continuum pressure after subtracting the wall-law gauge reference	Pa
p_{g}	1D gauge pressure produced by the wall law	Pa
$p_{\mathrm{ratio}} := p_{\mathrm{ref}} + p_{\mathrm{g}}$	reference-consistent pressure used in ratios by Convention C.3	Pa
p_{ref}	fixed pressure reference recorded by the ratio-output specification	Pa
$\pi_{\Delta p} = (z_{\mathrm{prox}}, z_{\mathrm{dist}})$	pressure-drop tap specification for the base 1D output map	m, m
π_{PR}	ratio-output specification $(z_{\mathrm{prox}}, z_{\mathrm{dist}}, \mathcal{T}_{\mathrm{HF}}, p_{\mathrm{ref}})$	mixed
$\langle f \rangle_{\mathcal{T}_{\mathrm{HF}}}$	time average of an integrable trace over the declared high-flow window; not the Appendix B inner product	same as f
$p_{\mathrm{out,g}}$	prescribed outlet gauge pressure in the same reference frame as p_{g}	Pa
$\Delta p(t)$	proximal-minus-distal pressure drop computed from p_{g}	Pa
$\overline{\Delta p}_{\mathcal{T}_{\mathrm{HF}}}$	scalar pressure-drop output averaged over recorded high-flow window $\mathcal{T}_{\mathrm{HF}}$	Pa
FFR-like shorthand	optional shorthand for PR_{1D} only when Convention C.3 and admissibility requirements are met	–
$\mathrm{PR}_{1D}(\pi_{\mathrm{PR}})$	schematic nonclinical pressure-ratio quantity under the same convention	–
τ_w or $\lvert\tau_w\rvert$	resolved traction-based WSS magnitude in 3D contexts	Pa
$\tau_{w,z}^{1D}$	signed 1D axial shear proxy from Q, A, and the recorded wall-friction/rheology closure	Pa
`shear_rate_1_s`	CSV output field for the implementation shear-rate proxy $\dot{\gamma}_{1D}$	s^{-1}
`nu_eff_cm2_s`	CSV output field for the effective kinematic viscosity used by the recorded rheology closure	cm^2/s
`rheology`	CSV and summary output field naming the recorded rheology descriptor	–

Remark C.4 (Viscosity notation). We write η for dynamic viscosity and

$$\nu := \frac{\eta}{\rho}$$

for kinematic viscosity. The distinction is dimensional: this report follows the convention that the coefficient in the Cauchy stress tensor is a dynamic viscosity: $D(\boldsymbol{u})$ has units of inverse time, so $\eta D(\boldsymbol{u})$ has pressure units. For generalized Newtonian models, the scalar viscosity function multiplying $2D(\boldsymbol{u})$ in the extra stress is likewise an effective dynamic viscosity, denoted

$$\eta_{\text{eff}}(\dot{\gamma}), \qquad \dot{\gamma} := \sqrt{2\boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u})}.$$

Here "this quantity" means the stress-level viscosity coefficient, not the divided-by-density coefficient. Several sources write the generalized viscosity function as $\mu(|D|^2)$, while related hemodynamics sources also use μ for viscosity coefficients in reduced and continuum settings [1, Sec. 3.1, p. 35, Sec. 3.6, p. 215] [23, Sec. 1.1, pp. 6–7, Eqs. (1.2)–(1.4), Fig. 4.4, p. 18]. To keep this report unambiguous, ν is always kinematic viscosity, while η and η_{eff} are dynamic viscosities.

D Mathematical Foundations

This appendix supplies citation anchors for the analytic notation used in the main chapters and in Appendix B. It separates domain and boundary language, classical function spaces, field notation, differential operators, and norm conventions so later sections can cite only the convention they need.

Convention D.1 (Standing regularity convention)**.** Unless a section states a stronger or weaker hypothesis, classical identities are used only under the regularity needed for their terms to be defined. In particular, fields have the displayed classical derivatives in the indicated domains, boundary traces are defined when boundary terms are written, and integration-by-parts or adjoint identities are invoked only when the required boundary terms are meaningful. When a weak formulation is discussed, the relevant Sobolev space, trace theorem, and boundary regularity assumptions are stated explicitly; Sobolev traces are not inferred from this classical convention. At minimum, boundaries are assumed piecewise C^1 when classical boundary integrals or integration-by-parts identities are used. Outward unit normals are then interpreted on the regular boundary pieces, hence surface-a.e. When a pointwise normal derivative is asserted on all of $\partial\Omega$, the boundary is assumed at least C^1.

Definition D.2 (Domain and boundary notation)**.** Let $\Omega \subset \mathbb{R}^n$ be a bounded open connected set, with $n \in \mathbb{N}$, and denote its closure by $\overline{\Omega}$. In the hemodynamics settings considered in this report, only $n \in [3]$ is used. The standing regularity convention in Convention D.1 applies whenever traces, outward unit normals, or integration by parts are used. For open Ω,

$$\overline{\Omega} = \Omega \cup \partial\Omega, \qquad \partial\Omega = \overline{\Omega} \setminus \Omega.$$

Since Ω is bounded, $\overline{\Omega}$ is closed and bounded and is therefore compact by the Heine–Borel theorem. The open set Ω itself need not be compact. Since Ω is bounded, $\partial\Omega$ is also compact. When the standing regularity convention supplies a piecewise C^1 boundary, the outward unit normal $\hat{\boldsymbol{n}}$ is interpreted on the regular boundary pieces, hence $dS_{\boldsymbol{x}}$-a.e. on $\partial\Omega$.

Definition D.3 (Time intervals and compact space-time sets)**.** For a final time $T_{\text{final}} > 0$, this report uses

$$I_T \;:=\; (0, T_{\text{final}}), \qquad \overline{I}_T \;:=\; [0, T_{\text{final}}].$$

The open interval I_T is used for interior-time differential identities and space-time PDE regions, while $\overline{I}_T$ is used when initial-time data, endpoint data, sampled time grids, or flow maps are included. When a local final time such as T_* is stated, the same convention applies with T_{final} replaced by that local final time.

For one-dimensional output fields, a compact space-time rectangle is written as

$$K \;:=\; [0, \ell] \times [0, T_{\text{final}}]$$

or by an explicitly stated local variant. Spaces such as $C(K)$ and $C^1([0,\ell])$ mean continuous functions on the named compact set, with the supremum norm when a norm is written. Spaces such as $L^p(K)$ use the product Lebesgue measure and the same almost-everywhere convention as Definition D.7.

Definition D.4 (Field and multi-index notation)**.** For $\Omega \subset \mathbb{R}^n$, we use the following field notation:

$$\begin{aligned}
&\phi : \Omega \to \mathbb{R} \ \text{ is a } \textit{scalar field},\\
&\boldsymbol{f} : \Omega \to \mathbb{R}^n \ \text{ is a } \textit{vector field},\\
&\boldsymbol{T} : \Omega \to \mathbb{R}^{n\times n} \ \text{ is a } \textit{(second-order) tensor field},
\end{aligned}$$

and we write $\phi(\boldsymbol{x})$, $\boldsymbol{f}(\boldsymbol{x})$, and $\boldsymbol{T}(\boldsymbol{x})$[1] for the values at a point $\boldsymbol{x} \in \Omega$. When referring to a physical quantity, its measurement units are indicated as $\left[\ \cdot\ \right]$, whereas dimensionless quantities are indicated as $\left[\ -\ \right]$. The k-th partial derivative of ϕ with respect to coordinate x_i is

$$\partial^k_{x_i}\phi \;\equiv\; \frac{\partial^k \phi}{\partial x_i^k}.$$

For a multi-index $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$, set $|\alpha| = \alpha_1 + \cdots + \alpha_n$. When the derivative exists classically, the corresponding derivative of a scalar field ϕ is

$$D^\alpha \phi \;\equiv\; \frac{\partial^{|\alpha|}\phi}{\partial x_1^{\alpha_1}\,\partial x_2^{\alpha_2}\,\dots\,\partial x_n^{\alpha_n}}.$$

For $|\alpha| = 1$, $D^\alpha\phi$ is a first-order partial derivative of ϕ, e.g. $\partial_{x_i}\phi$ where $\alpha_i = 1$ and $\alpha_j = 0$ for all $j \in [n]\setminus\{i\}$.

Definition D.5 (Classical function-space notation)**.** Given a domain Ω and a scalar field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, a scalar function space on Ω is an $\mathbb{F}$-vector subspace of the maps $f : \Omega \to \mathbb{F}$. Unless a complex-valued space is explicitly specified, this report uses real-valued spaces. Thus $C^0(\Omega)$ denotes the real-valued continuous functions on the open domain Ω. For $k \in \mathbb{N}_0$, $C^k(\Omega)$ denotes functions $\phi : \Omega \to \mathbb{R}$ such that $D^\alpha\phi$ exists and is continuous in Ω for every $|\alpha| \le k$. By contrast, $C^k(\overline{\Omega})$ denotes functions in $C^k(\Omega)$ whose derivatives through order k extend continuously to the closure $\overline{\Omega}$. The space $C^\infty(\Omega)$ is the intersection of all $C^k(\Omega)$ for $k \in \mathbb{N}_0$, i.e. the infinitely differentiable functions on Ω. The notation $C^k(\Omega; \mathbb{R}^m)$ denotes the maps $\boldsymbol{f} : \Omega \to \mathbb{R}^m$ with components in $C^k(\Omega)$, i.e., $\boldsymbol{f} = (f_i)_{i\in[m]}$ with $f_i \in C^k(\Omega)$ for all $i \in [m]$.

Definition D.6 (Differential-operator conventions)**.** For real-valued $\phi \in C^1(\Omega)$, the partial derivative with respect to coordinate x_i is

$$\partial_{x_i}\phi = \frac{\partial \phi}{\partial x_i}.$$

[1] A second-order tensor value $\boldsymbol{T}(\boldsymbol{x}) \in \mathbb{R}^{n\times n}$ can be identified with the matrix of a linear map $\mathbb{R}^n \to \mathbb{R}^n$ acting on vectors by matrix-vector multiplication.

Named-coordinate partial derivatives use the same compact convention; for example,

$$\partial_t \phi = \frac{\partial \phi}{\partial t}, \qquad \partial_z f = \frac{\partial f}{\partial z}, \qquad \partial_A p = \frac{\partial p}{\partial A}.$$

Cylindrical derivatives such as $\partial_r u$ and $\partial_\theta u$ are read analogously. Boundary notation retains ∂, as in $\partial V(t)$ and $\partial\Omega$. For $\boldsymbol{x} \in \Omega$ and a direction $\boldsymbol{v} \in \mathbb{R}^n$, the directional derivative is

$$D_{\boldsymbol{v}}\phi(\boldsymbol{x}) \;:=\; \langle \nabla\phi(\boldsymbol{x}),\; \boldsymbol{v}\rangle_{\mathbb{R}^n}.$$

If $\phi \in C^1(\overline{\Omega})$, $\boldsymbol{x} \in \partial\Omega$, and $\hat{\boldsymbol{n}}$ is the outward unit normal at $\boldsymbol{x}$, then the boundary normal derivative is

$$\partial_{\hat{\boldsymbol{n}}}\phi(\boldsymbol{x}) \;:=\; \langle \nabla\phi(\boldsymbol{x}),\; \hat{\boldsymbol{n}}\rangle_{\mathbb{R}^n}.$$

For Lebesgue measure $d\boldsymbol{x}$ on $\mathbb{R}^n$ the integral of ϕ in Ω is

$$\int_\Omega \phi(\boldsymbol{x})\, d\boldsymbol{x}.$$

If $\partial\Omega$ is of class C^1 and $\phi \in C^0(\overline{\Omega})$, then the classical boundary trace $\phi|_{\partial\Omega}$ is continuous on $\partial\Omega$. Since Ω is bounded and $\partial\Omega$ is a compact C^1 hypersurface, this trace belongs to $L^1(\partial\Omega; dS_{\boldsymbol{x}})$. The corresponding boundary integral is written

$$\int_{\partial\Omega} \phi(\boldsymbol{x})\; dS_{\boldsymbol{x}}.$$

We use the column-gradient convention for scalar fields. For $\phi \in C^1(\Omega)$, the gradient is

$$\nabla\phi \;:=\; (\partial_{x_i}\phi)_{i\in[n]} \in \mathbb{R}^n.$$

For $\phi \in C^2(\Omega)$, the Laplacian is

$$\Delta\phi \;:=\; \nabla\cdot\nabla\phi \;=\; \sum_{i\in[n]} \partial_{x_i}^2\phi.$$

For vector-valued $\boldsymbol{f} \in C^1(\Omega;\mathbb{R}^n)$ with components $\boldsymbol{f} = (f_i)_{i\in[n]}$, the vector gradient is the Jacobian with entries $(\nabla\boldsymbol{f})_{ij} \;:=\; \partial_{x_j} f_i$. The divergence is the trace of this Jacobian:

$$\operatorname{div}\boldsymbol{f} \;:=\; \nabla\cdot\boldsymbol{f} \;=\; \sum_{i\in[n]} \partial_{x_i} f_i \;=\; \operatorname{tr}(\nabla\boldsymbol{f}).$$

For $\boldsymbol{f} \in C^2(\Omega;\mathbb{R}^n)$, the Laplacian is defined componentwise. The domain integral is also componentwise whenever the components are integrable:

$$\begin{aligned}\Delta \boldsymbol{f} &:= (\Delta f_i)_{i\in[n]}\\ \int_\Omega \boldsymbol{f}(\boldsymbol{x})\, d\boldsymbol{x} &:= \left(\int_\Omega f_i(\boldsymbol{x})\, d\boldsymbol{x}\right)_{i\in[n]}.\end{aligned}$$

For tensor-valued $\boldsymbol{T} \in C^1(\Omega;\mathbb{R}^{n\times n})$ with entries $\boldsymbol{T} = (T_{ij})_{i,j\in[n]}$, we use the row-divergence convention $(\mathbf{div}\boldsymbol{T})_i = \sum_{j\in[n]} \partial_{x_j} T_{ij}$. Equivalently, if $\boldsymbol{t}_i = (T_{ij})_{j\in[n]}$ denotes the i-th row of $\boldsymbol{T}$, then

$$\begin{aligned}\mathbf{div}\,\boldsymbol{T} &:= (\mathrm{div}\,\boldsymbol{t}_i)_{i\in[n]}\\ &= (\nabla\cdot\boldsymbol{t}_i)_{i\in[n]}\\ &= \left(\sum_{j\in[n]} \partial_{x_j} T_{ij}\right)_{i\in[n]}.\end{aligned}$$

For $\boldsymbol{T} \in C^2(\Omega;\mathbb{R}^{n\times n})$, the Laplacian is defined componentwise. The domain integral is also defined componentwise whenever the entries are integrable:

$$\begin{aligned}\Delta \boldsymbol{T} &:= (\Delta T_{ij})_{i,j\in[n]}\\ \int_\Omega \boldsymbol{T}(\boldsymbol{x})\, d\boldsymbol{x} &:= \left(\int_\Omega T_{ij}(\boldsymbol{x})\, d\boldsymbol{x}\right)_{i,j\in[n]}.\end{aligned}$$

Definition D.7 (C^k and L^p norm conventions)**.** Let $\Omega \subset \mathbb{R}^n$ be bounded and open. We write $C^k(\overline{\Omega})$ for the functions $\phi \in C^k(\Omega)$ such that each derivative $D^\alpha\phi$, $|\alpha| \le k$, extends continuously to $\overline{\Omega}$. The norm is

$$\|\phi\|_{C^k(\overline{\Omega})} := \sum_{|\alpha|\le k} \sup_{\boldsymbol{x}\in\overline{\Omega}} |D^\alpha \phi(\boldsymbol{x})|.$$

With this norm, $C^k(\overline{\Omega})$ is a Banach space: every Cauchy sequence in $C^k(\overline{\Omega})$ converges, with respect to $\|\cdot\|_{C^k(\overline{\Omega})}$, to a limit in $C^k(\overline{\Omega})$. In particular, for $k = 0$, we have

$$\|\phi\|_{C^0(\overline{\Omega})} := \sup_{\boldsymbol{x}\in\overline{\Omega}} |\phi(\boldsymbol{x})| = \max_{\boldsymbol{x}\in\overline{\Omega}} |\phi(\boldsymbol{x})|\,.$$

For continuous functions on compact sets, the max norm agrees with the usual supremum norm.

For $1 \le p < \infty$, $L^p(\Omega)$ denotes the space of equivalence classes of real-valued measurable functions on Ω, identified up to equality almost everywhere, with finite norm

$$\|\phi\|_{L^p(\Omega)} := \left(\int_\Omega |\phi(\boldsymbol{x})|^p\, d\boldsymbol{x}\right)^{1/p}.$$

For $p = \infty$,

$$\|\phi\|_{L^\infty(\Omega)} \;:=\; \operatorname*{ess\,sup}_{\boldsymbol{x}\in\Omega} |\phi(\boldsymbol{x})|.$$

With this almost-everywhere identification, $L^p(\Omega)$ is a Banach space for $1 \le p \le \infty$.

The notation $L^p(\Omega; \mathbb{R}^m)$ denotes equivalence classes of measurable maps $\boldsymbol{f} : \Omega \to \mathbb{R}^m$, identified up to equality a.e., whose components belong to $L^p(\Omega)$, i.e., $\boldsymbol{f} = (f_i)_{i\in[m]}$ with $f_i \in L^p(\Omega)$ for all $i \in [m]$. When a norm on $L^p(\Omega; \mathbb{R}^m)$ is needed, we use the Euclidean norm in the range. For $1 \le p < \infty$,

$$\|\boldsymbol{f}\|_{L^p(\Omega;\mathbb{R}^m)} \;:=\; \left(\int_\Omega \|\boldsymbol{f}(\boldsymbol{x})\|_{\mathbb{R}^m}^p \, d\boldsymbol{x}\right)^{1/p}.$$

For $p = \infty$, the corresponding convention is

$$\|\boldsymbol{f}\|_{L^\infty(\Omega;\mathbb{R}^m)} \;:=\; \operatorname*{ess\,sup}_{\boldsymbol{x}\in\Omega} \|\boldsymbol{f}(\boldsymbol{x})\|_{\mathbb{R}^m}.$$

For matrix or second-order tensor values, the pointwise range norm is the Frobenius norm

$$\|\boldsymbol{T}\|_{\mathrm{F}} \;:=\; (\boldsymbol{T} : \boldsymbol{T})^{1/2} = \left(\sum_{i,j\in[n]} T_{ij}^2\right)^{1/2}.$$

Thus tensor-valued L^p norms are formed by replacing $|\phi(\boldsymbol{x})|$ in the scalar L^p formula with $\|\boldsymbol{T}(\boldsymbol{x})\|_{\mathrm{F}}$.

E Continuum Derivation Details

This appendix supplements the continuum mechanics details that support the main-body assumptions without interrupting the reduced-order forward-model narrative within this report. It is appendix background for notation and model hierarchy, not additional evidence for the first controlled 1D study. All identities below are classical identities under the regularity needed for the displayed derivatives, traces, and changes of variables. The standing regularity convention, boundary notation, and differential-operator convention are those of Convention D.1, Definition D.2, and Definition D.6. In particular, **div** is the row-divergence of a tensor field, and boundary forms assume the stated trace and normal hypotheses; no weak formulation is being asserted in this appendix.

E.1 Material Derivative and Trajectory Details

For a scalar field $\phi \in C^1(\mathcal{Q}_T)$ and a sufficiently regular flow map $\mathcal{L}_t$, define the Lagrangian pullback $\widehat{\phi}(t,\boldsymbol{\xi}) := \phi(t,\mathcal{L}_t(\boldsymbol{\xi}))$. Differentiating with respect to time along a trajectory and using $\partial_t\mathcal{L}_t(\boldsymbol{\xi}) = \boldsymbol{u}(t,\mathcal{L}_t(\boldsymbol{\xi}))$ gives

$$\frac{d}{dt}\phi(t,\mathcal{L}_t(\boldsymbol{\xi})) = \partial_t\phi(t,\boldsymbol{x}) + \nabla\phi(t,\boldsymbol{x})\cdot\boldsymbol{u}(t,\boldsymbol{x}), \qquad \boldsymbol{x} = \mathcal{L}_t(\boldsymbol{\xi}).$$

The identity

$$\mathrm{div}(\phi\boldsymbol{u}) = \nabla\phi\cdot\boldsymbol{u} + \phi\mathrm{div}\boldsymbol{u}$$

also gives

$$D_t\phi = \partial_t\phi + \mathrm{div}(\phi\boldsymbol{u}) - \phi\mathrm{div}\boldsymbol{u}.$$

E.2 Jacobian Evolution and Reynolds Transport

Let $\widehat{\boldsymbol{F}}_t = \nabla_{\boldsymbol{\xi}}\mathcal{L}_t$ and $\widehat{J}_t = \det\widehat{\boldsymbol{F}}_t$. Here $\mathcal{L}_t$ is assumed to be an orientation-preserving C^1 change of variables on the material volume being considered, with the additional time regularity needed to differentiate $\widehat{\boldsymbol{F}}_t$ and $\widehat{J}_t$. The standard Jacobi identity gives

$$\partial_t\widehat{J}_t = \widehat{J}_t\,\mathrm{tr}\Big(\partial_t\widehat{\boldsymbol{F}}_t\,\widehat{\boldsymbol{F}}_t^{-1}\Big) = \widehat{J}_t\,\mathrm{div}\boldsymbol{u}(t,\mathcal{L}_t(\boldsymbol{\xi})),$$

which is the Lagrangian form of $D_tJ_t = J_t\mathrm{div}\boldsymbol{u}$ [2, Part I, Ch. 1, Sec. 1.3].

For $V(t) = \mathcal{L}_t(V(0))$, change variables in $I(t) = \int_{V(t)} \phi(t, \boldsymbol{x})\, d\boldsymbol{x}$:

$$I(t) = \int_{V(0)} \phi(t, \mathcal{L}_t(\boldsymbol{\xi}))\widehat{J}_t(\boldsymbol{\xi})\, d\boldsymbol{\xi}.$$

Differentiating under the integral sign and substituting the Jacobian-evolution identity gives

$$I'(t) = \int_{V(t)} \left(D_t\phi + \phi \operatorname{div}\boldsymbol{u}\right)\, d\boldsymbol{x} = \int_{V(t)} \left(\partial_t\phi + \operatorname{div}(\phi\boldsymbol{u})\right)\, d\boldsymbol{x}.$$

If $V(t)$ has Lipschitz boundary and the trace of $\phi\boldsymbol{u}$ is integrable on $\partial V(t)$, the divergence theorem gives the boundary form

$$I'(t) = \int_{V(t)} \partial_t\phi\; d\boldsymbol{x} + \int_{\partial V(t)} \phi\, \boldsymbol{u} \cdot \hat{\boldsymbol{n}}\, dS.$$

E.3 Linear Momentum Balance Details

Starting from the material-volume balance

$$\frac{d}{dt}\int_{V(t)} \rho\boldsymbol{u}\, dV = \int_{\partial V(t)} \boldsymbol{T}\hat{\boldsymbol{n}}\, dS + \int_{V(t)} \rho\boldsymbol{f_b}\, dV,$$

Reynolds transport gives

$$\frac{d}{dt}\int_{V(t)} \rho\boldsymbol{u}\, dV = \int_{V(t)} \left(D_t(\rho\boldsymbol{u}) + \rho\boldsymbol{u}\operatorname{div}\boldsymbol{u}\right) dV.$$

Using mass conservation, $D_t\rho + \rho\operatorname{div}\boldsymbol{u} = 0$, this reduces to

$$\int_{V(t)} \rho D_t\boldsymbol{u}\, dV = \int_{V(t)} \rho\left(\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) dV.$$

The surface traction term satisfies

$$\int_{\partial V(t)} \boldsymbol{T}\hat{\boldsymbol{n}}\, dS = \int_{V(t)} \mathbf{div}(\boldsymbol{T})\, dV.$$

This is the tensor row-divergence convention from Definition D.6, applied under the regularity needed for the stress trace $\boldsymbol{T}\hat{\boldsymbol{n}}$ and the volume integral of $\mathbf{div}\boldsymbol{T}$. Since $V(t)$ is arbitrary within the localization class, the differential form is

$$\rho\left(\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = \mathbf{div}(\boldsymbol{T}) + \rho\boldsymbol{f_b}.$$

The conservative Eulerian form

$$\partial_t(\rho\boldsymbol{u}) + \mathbf{div}(\rho\boldsymbol{u}\otimes\boldsymbol{u}) = \mathbf{div}(\boldsymbol{T}) + \rho\boldsymbol{f_b}$$

is equivalent whenever the continuity equation holds, because

$$\partial_t(\rho\boldsymbol{u}) + \mathbf{div}(\rho\boldsymbol{u}\otimes\boldsymbol{u}) = \rho\left(\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) + \boldsymbol{u}\left(\partial_t\rho + \operatorname{div}(\rho\boldsymbol{u})\right).$$

F Navier–Stokes Coordinate and Energy Details

This appendix collects mathematical orientation material for the 3D Navier–Stokes model. It is background for the continuum hierarchy; it is not evidence for the 1D study specification or any downstream validation claim. All formulas are read under the standing regularity and differential-operator conventions in Appendix D. In particular, $\mathbf{div}$ is row-divergence for tensor fields, $\Delta \boldsymbol{u}$ is the componentwise vector Laplacian, tensor norms are Frobenius norms when written, and η is dynamic viscosity while $\nu = \eta/\rho$ is kinematic viscosity as in Remark C.4.

F.1 Viscous-Term Simplification and Energy Context

For a Newtonian fluid with constant dynamic viscosity η and sufficiently smooth velocity field,

$$\mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \eta \mathbf{div}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top) = \eta \Delta \boldsymbol{u} + \eta \nabla(\operatorname{div} \boldsymbol{u}).$$

For incompressible flow, $\operatorname{div} \boldsymbol{u} = 0$, so

$$\mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \eta \Delta \boldsymbol{u}, \qquad \rho^{-1} \mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \nu \Delta \boldsymbol{u}, \qquad \nu = \eta/\rho.$$

This term is the momentum-diffusion contribution; it does not disappear in the incompressible Newtonian limit.

Classical local well-posedness results for Navier–Stokes depend on the domain, boundary conditions, forcing, initial-data space, and solution class. Under sufficiently smooth compatible data one obtains a unique strong solution on a maximal time interval. If the maximal interval is finite, continuation fails through blow-up of the norm controlled by the chosen well-posedness theory.

F.2 Cylindrical Coordinate Form

For (r, θ, z) with orthonormal basis $(\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_z)$,

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad \boldsymbol{u} = u_r \boldsymbol{e}_r + u_\theta \boldsymbol{e}_\theta + u_z \boldsymbol{e}_z.$$

The component formulas are local cylindrical-coordinate formulas on patches where $r > 0$. Axis regularity, wall traces, and boundary conditions require separate hypotheses and are not supplied by the coordinate notation alone. The differential operators used in cylindrical divided-by-density Navier–Stokes forms are

$$
\begin{aligned}
(\boldsymbol{u}\cdot\nabla) &= u_r\partial_r + \frac{u_\theta}{r}\partial_\theta + u_z\partial_z,\\
\nabla p &= \partial_r p\,\boldsymbol{e}_r + \frac{1}{r}\partial_\theta p\,\boldsymbol{e}_\theta + \partial_z p\,\boldsymbol{e}_z,\\
\mathrm{div}\boldsymbol{u} &= \frac{1}{r}\partial_r(ru_r) + \frac{1}{r}\partial_\theta u_\theta + \partial_z u_z.
\end{aligned}
$$

The rate-of-deformation components in the cylindrical orthonormal basis are

$$
\begin{aligned}
&D_{rr} = \partial_r u_r, && D_{\theta\theta} = \frac{1}{r}\partial_\theta u_\theta + \frac{u_r}{r}, && D_{zz} = \partial_z u_z,\\
&D_{r\theta} = \frac{1}{2}\left(\partial_r u_\theta - \frac{u_\theta}{r} + \frac{1}{r}\partial_\theta u_r\right), && D_{rz} = \frac{1}{2}(\partial_r u_z + \partial_z u_r), && D_{\theta z} = \frac{1}{2}\left(\frac{1}{r}\partial_\theta u_z + \partial_z u_\theta\right).
\end{aligned}
$$

Use the scalar shear-rate convention $\dot\gamma = \sqrt{2\boldsymbol{D}(\boldsymbol{u}):\boldsymbol{D}(\boldsymbol{u})}$. Then

$$
\tau_{ij} = 2\eta_{\mathrm{eff}}(\dot\gamma)D_{ij}, \qquad i,j\in\{r,\theta,z\}.
$$

When a source writes the law as a function of $D:D$ or $|D|_{\mathrm{F}}^2$, this report translates it through this convention, consistent with Remark C.4. Writing $\boldsymbol{f} = f_r\boldsymbol{e}_r + f_\theta\boldsymbol{e}_\theta + f_z\boldsymbol{e}_z$, the momentum components are

$$
\begin{aligned}
\text{(radial)}\quad & \partial_t u_r + u_r\partial_r u_r + \frac{u_\theta}{r}\partial_\theta u_r + u_z\partial_z u_r - \frac{u_\theta^2}{r}\\
&= -\frac{1}{\rho}\partial_r p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{rr}) + \frac{1}{r}\partial_\theta\tau_{r\theta} + \partial_z\tau_{rz} - \frac{1}{r}\tau_{\theta\theta}\right] + f_r,\\
\text{(azimuthal)}\quad & \partial_t u_\theta + u_r\partial_r u_\theta + \frac{u_\theta}{r}\partial_\theta u_\theta + u_z\partial_z u_\theta + \frac{u_r u_\theta}{r}\\
&= -\frac{1}{\rho r}\partial_\theta p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{r\theta}) + \frac{1}{r}\partial_\theta\tau_{\theta\theta} + \partial_z\tau_{\theta z} + \frac{1}{r}\tau_{r\theta}\right] + f_\theta,\\
\text{(axial)}\quad & \partial_t u_z + u_r\partial_r u_z + \frac{u_\theta}{r}\partial_\theta u_z + u_z\partial_z u_z\\
&= -\frac{1}{\rho}\partial_z p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{rz}) + \frac{1}{r}\partial_\theta\tau_{\theta z} + \partial_z\tau_{zz}\right] + f_z.
\end{aligned}
$$

The incompressibility condition is

$$
\frac{1}{r}\partial_r(ru_r) + \frac{1}{r}\partial_\theta u_\theta + \partial_z u_z = 0.
$$

G Numerical Methods Details

This appendix supplies the numerical-method detail behind Sec. 1.2.3 and the comparison boundary in Sec. 1.2.4. It is supporting method detail, not a claim that the current solver has external-solver parity. The continuous identities are read under Convention D.1; computed quantities are read through the following numerical-record convention.

Convention G.1 (Computed realization record)**.** A computed realization is interpreted only through the model, closure, discretization, grid, timestep policy, boundary-state rule, output operator, and diagnostic metadata recorded for that realization. Symbols such as D_t, D_z, numerical fluxes, residual arrays, and sampled norms denote recorded discrete objects unless a section explicitly switches back to a continuum identity.

The time interval, output rectangle, and norm notation follow Definition D.3 and Definition D.7; discrete residuals use the recorded D_t, D_z, grid, and quadrature weights rather than unnamed continuum derivatives.

G.1 Elastic-potential balance-law form

The reduced state is $\mathbf{U} = (A, Q)^\top$. For the $\mathcal{W}_{\mathrm{elastic1D}}$ balance-law contract with prescribed momentum-flux correction α, the reference elastic-potential form is

$$\begin{aligned} \partial_t A + \partial_z Q &= 0, \\ \partial_t Q + \partial_z \left(\alpha \frac{Q^2}{A} + \Psi(A, z, t) \right) &= S_{\mathrm{geom}}(A, z, t) - C_f(z, A, Q) \frac{Q}{A}. \end{aligned}$$

The elastic potential satisfies

$$\partial_A \Psi(A, z, t) = \frac{A}{\rho} \partial_A p_{\mathrm{g}}(A, z, t),$$

and the geometry source is

$$S_{\mathrm{geom}}(A, z, t) = \partial_z \Psi(A, z, t)\big|_A - \frac{A}{\rho}\, \partial_z p_{\mathrm{g}}(A, z, t)\big|_A .$$

For the baseline wall law, assuming p_{ext} is independent of A,

$$c^2(A, z) = \frac{A}{\rho} \partial_A p_{\mathrm{g}}(A, z, t) = \frac{\beta(z)\sqrt{A}}{2\rho A_0(z)}.$$

The strictly hyperbolic positive-area region for this classical model has wave speeds

$$\lambda_\pm(A,Q;z)=\alpha\frac{Q}{A}\pm\sqrt{c^2(A,z)+\alpha(\alpha-1)\left(\frac{Q}{A}\right)^2}.$$

These reduce to $Q/A \pm c(A,z)$ in the special $\alpha = 1$ reference case. The boundary-state construction is intended for the subcritical case $\lambda_- < 0 < \lambda_+$, so one datum enters at each end of the vessel. The displayed derivatives of A_0, β, p_g, and Ψ are continuum notation for the model contract. A computed record may realize them through explicitly recorded finite-difference or source-split operators; changing that realization changes the computed map.

G.2 Current semi-discrete solver operator

The current Canic extended implementation uses the code variable A for a radius-squared area coordinate. In this subsection write that coordinate as $a = R^2$. The physical circular cross-sectional area is $A_\mathrm{phys} = \pi a$. This is a local convention for the active solver record; the continuum and classical 1D model sections keep A for physical cross-sectional area.

For N finite-volume cells, let

$$z_i=\left(i-\frac{1}{2}\right)\Delta z,\qquad \Delta z=\frac{L}{N},\qquad u_h(t)=\left(a_1(t),\ldots,a_N(t),Q_1(t),\ldots,Q_N(t)\right)^\top.$$

The semi-discrete system exposed by `semidiscretize` and `rhs!` is

$$\dot{u}_h=\mathcal{L}_h(u_h,t;\theta),$$

where θ denotes the recorded physical, geometry, grid, and solver parameters. Componentwise,

$$\begin{aligned}\dot{a}_i&=-\frac{\widehat{F}^a_{i+1/2}-\widehat{F}^a_{i-1/2}}{\Delta z},\\ \dot{Q}_i&=-\frac{\widehat{F}^Q_{i+1/2}-\widehat{F}^Q_{i-1/2}}{\Delta z}+\widehat{S}_i.\end{aligned}$$

All divisions by a in the implementation use the recorded positive-area floor.

With $K = Eh/(1-\sigma^2)$, velocity-profile momentum factor $\alpha_\mathcal{P}$, and $\alpha_c(R_0') = -2(R_0')^2/35$, the current full-state flux is

$$\mathbf{F}(a,Q,z)=\left(Q,\,(\alpha_\mathcal{P}+\alpha_c(R_0'(z)))\,\frac{Q^2}{a}+\frac{K}{3\rho R_\mathrm{max}^2}a^{3/2}\right)^\top.$$

At each interface,

$$\widehat{\mathbf{F}}_{i+1/2} = \frac{1}{2}\left[\mathbf{F}(\mathbf{U}^-_{i+1/2}, z_{i+1/2}) + \mathbf{F}(\mathbf{U}^+_{i+1/2}, z_{i+1/2})\right] - \frac{1}{2}s_{i+1/2}\left(\mathbf{U}^+_{i+1/2} - \mathbf{U}^-_{i+1/2}\right),$$

where $\mathbf{U} = (a, Q)^\top$ and the present implementation uses neighboring first-order states plus ghost boundary states. The recorded local speed is

$$s_{i+1/2} = \max\left\{s_h(\mathbf{U}^-_{i+1/2}, z_{i+1/2}), s_h(\mathbf{U}^+_{i+1/2}, z_{i+1/2})\right\},$$

with

$$\begin{aligned} s_h(a, Q, z) &= \max\left\{\left|\alpha_{\text{eff}} u - \chi\right|, \left|\alpha_{\text{eff}} u + \chi\right|\right\}, \\ \alpha_{\text{eff}} &= \alpha_{\mathcal{P}} + \alpha_c(R_0'(z)), \qquad u = \frac{Q}{a}, \\ \chi^2 &= (\alpha_{\text{eff}} u)^2 - \alpha_{\text{eff}} u^2 + \frac{K}{2\rho R_{\max}^2}\sqrt{a}. \end{aligned}$$

The source term uses the recorded centered interior and one-sided boundary difference D^h_z on the cell averages. Let $g_{\mathcal{P}}$ denote the profile shear factor,

$$\dot\gamma_{1D,i} = g_{\mathcal{P}}\frac{|Q_i|}{a_i\sqrt{a_i}}, \qquad \nu^{\mathcal{R}}_{\text{eff},i} = \frac{\eta^{\mathcal{R}}_{\text{eff}}(\dot\gamma_{1D,i})}{\rho},$$

and $\alpha_c'(z) = -4R_0'(z)R_0''(z)/35$. The implemented source is

$$\begin{aligned} \widehat{S}_i &= -2\nu^{\mathcal{R}}_{\text{eff},i} g_{\mathcal{P}}\frac{Q_i}{a_i} + \frac{K}{\rho R_{\max}^2}a_i R_0'(z_i) - a_i\,\widehat{\partial_z p_2}_i + \frac{Q_i^2}{a_i}\alpha_c'(z_i), \\ \widehat{\partial_z p_2}_i &= \nu^{\mathcal{R}}_{\text{eff},i} g_{\mathcal{P}}\left[\frac{Q_i R_0''(z_i)}{a_i R_0(z_i)} - \frac{Q_i (R_0'(z_i))^2}{a_i R_0(z_i)^2} + \frac{(D^h_z Q)_i R_0'(z_i)}{a_i R_0(z_i)} - \frac{Q_i (D^h_z a)_i R_0'(z_i)}{a_i^2 R_0(z_i)}\right]. \end{aligned}$$

For the `newtonian` descriptor, this reduces to the constant value $\nu^{\mathcal{R}}_{\text{eff},i} = \nu$. For the non-Newtonian closures, the code applies the recorded shear-rate floor and optional minimum/maximum dynamic viscosity clamps before dividing by density. For the power-family profile, $g_{\mathcal{P}} = \gamma + 2 = \alpha_{\mathcal{P}}/(\alpha_{\mathcal{P}} - 1)$. The flat profile is not evaluated through that singular limit; it records $\alpha_{\mathcal{P}} = 1$ and an explicit finite $g_{\mathcal{P}}$.

G.3 Time and backend realization

The native backend advances the semi-discrete operator with the recorded third-order strong-stability-preserving Runge–Kutta map

$$u_h^{n+1} = \Phi^{\text{SSPRK3}}_{\Delta t_n}\left(u_h^n; \mathcal{L}_h\right).$$

In expanded form, using Π for the recorded positive-area projection on the a-components,

$$\begin{aligned}
u^{(1)} &= \Pi\left(u^n + \Delta t_n \mathcal{L}_h(u^n, t^n)\right), \\
u^{(2)} &= \Pi\left(\frac{3}{4}u^n + \frac{1}{4}\left[u^{(1)} + \Delta t_n \mathcal{L}_h(u^{(1)}, t^n + \Delta t_n)\right]\right), \\
u^{n+1} &= \Pi\left(\frac{1}{3}u^n + \frac{2}{3}\left[u^{(2)} + \Delta t_n \mathcal{L}_h(u^{(2)}, t^n + \Delta t_n)\right]\right).
\end{aligned}$$

The native step size is

$$\Delta t_n = \min\left\{\Delta t_{\max}, \frac{\mathrm{CFL}\,\Delta z}{\max_i s_h(a_i^n, Q_i^n, z_i)}, T_{\mathrm{final}} - t^n\right\}.$$

The SciML backend constructs an `ODEProblem` for the same $\dot{u}_h = \mathcal{L}_h(u_h, t; \theta)$ and applies the selected OrdinaryDiffEq policy recorded in the solve specification. Thus backend comparisons change the time integrator, not the semi-discrete spatial operator.

G.4 Boundary-state realization

The current finite-volume record imposes one incoming datum at each boundary in the subcritical regime using the code's Riemann-invariant construction. Let

$$c_0 = \left(\frac{K}{2\rho R_{\max}^2}\right)^{1/2}.$$

At the inlet, the incoming flow is $Q_{\mathrm{in}} = R_0(0)^2 \bar{u}_{\mathrm{in}}$. The outgoing invariant from the first interior cell is

$$w_- = \frac{Q_1}{a_1} - 4c_0 a_1^{1/4},$$

and the ghost coordinate a_{in} solves

$$\frac{Q_{\mathrm{in}}}{a_{\mathrm{in}}} - 4c_0 a_{\mathrm{in}}^{1/4} = w_-.$$

At the outlet, the radius-squared coordinate is fixed to $a_{\mathrm{out}} = R_0(L)^2$. The outgoing invariant from the last interior cell is

$$w_+ = \frac{Q_N}{a_N} + 4c_0 a_N^{1/4},$$

and the ghost flow is

$$Q_{\mathrm{out}} = a_{\mathrm{out}} w_+ - 4c_0 a_{\mathrm{out}}^{5/4}.$$

The boundary update therefore constrains the incoming information and leaves the outgoing information tied to the interior state. Boundary-quality plots and reflected-wave sidecars are diagnostics until explicit thresholds are admitted.

G.5 Residual diagnostics

Stored residuals are post-processing diagnostics on sampled fields, not proof of the continuous PDE residual. The symbols D_t and D_z in this subsection denote the recorded temporal and spatial finite-difference operators on the saved grid, not weak derivatives. For a stored field sequence, the semi-discrete defect is read as

$$\mathcal{E}_h^n = D_t u_h^n - \mathcal{L}_h(u_h^n, t^n; \theta).$$

Componentwise, this gives

$$\mathcal{E}_i^a(t^n) = D_t a_i^n - \mathcal{L}_{h,i}^a(u_h^n, t^n; \theta), \qquad \mathcal{E}_i^Q(t^n) = D_t Q_i^n - \mathcal{L}_{h,i}^Q(u_h^n, t^n; \theta).$$

The E1 diagnostic summary reports these residuals on reviewed interior cells with recorded finite-field, positivity, CFL, and pressure-drop sign checks. The numbers support the dataset contract and local consistency reading; they do not establish a convergence theorem or external parity. Any residual norm or error norm reported from these arrays is a sampled norm tied to the numerical record, rather than an unqualified continuum $L^p(K)$ norm.

H Code Availability and Build Statement

This report is built from the local LaTeX sources rooted at `final-report.tex`. The compiled PDF should be treated as the rendered form of those sources. The local checkout used for this report is rooted at `/Users/doe/hemodynamics/masters-report`. The parent checkout for this report-source update was git commit `019fa39`. No git remote is configured in this checkout, so this appendix does not claim a public repository or archival URL.

The Section 2 comparison record is the ignored local directory `simulations/output/3d_comparison/full_t1_native_nx400/`. The command shape for regenerating the available comparison, when the ignored raw 3D inputs are restored, is:

```
./scripts/julia-release -e 'include("simulations/canic_extended_1d/CanicExtended1D.jl");
    using .CanicExtended1D; run_available_resolved3d_comparison(output_dir="simulations/
    output/3d_comparison/full_t1_native_nx400", overwrite=true)'
```

The raw upstream XDMF/HDF5 inputs under `simulations/data/3d/canic_case3/` are ignored by git and are absent from this checkout. Therefore the comparison is not independently reproducible from the tracked report package alone. The tracked static data copied into the PDF has the following SHA-256 hashes:

Tracked file	SHA-256
figures/static/static/data/stenosis-comparison/section-mean.dat	5939c015eaa848dbd62e0531b26799a3297eb3ebb7c936fd6ce66e9f4b8e649e
figures/static/static/data/stenosis-comparison/radial-case77.dat	90d4766201f5610efdd41d93f0717a6c04485a00f4f60a272ba46207c7065f8a
figures/static/static/data/stenosis-comparison/radial-case60.dat	e35b4f1e9016e22ba1ee9feb6389452732eb81fb0ea7a9d4a686f64d89e1bbbc

The ignored local comparison record observed in this checkout contains the following output hashes; these files are local evidence, not tracked archive contents:

Ignored record file	SHA-256
comparison_summary.csv	8dbb4eb5f6e9fa369fa1a4bf215ff2c3 80b788d2b9b89f828718ae1485e79b6b
section_mean_case77_sev23.csv	4efafa3c314eaa19193f3c884c9b9607 9423a2bf2985cde4cc6511f0dfaf9348
section_mean_case60_sev40.csv	48a6af7ef5f50bdadfcf7b043d7632acc fc73af41750e93f02e701c16fb948b4
radial_profile_case77_sev23.csv	29bb3e13fa34396c27c91d0183bcac1 16516bedcc5b92be33cb832126216890d
radial_profile_case60_sev40.csv	994091f5e74d39249d8d7b9ec374d07 b36eec9fdef72050bc712f5615443458c
section_mean_overlay.svg	c6ac9de7f3cc92782c61753c346e66e 405fc6f3ab22d45e435c7afa5d41aa322
radial_profile_case77_sev23.svg	fb60c63ef2963c6f13f8e28c6d0b306 485326318e32f83d12c4f296652834c63
radial_profile_case60_sev40.svg	e8a3794c1e07d3ea58ffeb59be3c0558 e9009473cf3fb304a8b74aa1250f7bf7

References

[1] T. Köppl and R. Helmig. *Dimension Reduced Modeling of Blood Flow in Large Arteries. An Introduction for Master Students and First Year Doctoral Students*. Mathematical Engineering. Springer Cham, 2023. DOI: 10.1007/978-3-031-33087-2.

[2] G. P. Galdi, A. M. Robertson, R. Rannacher, and S. Turek. *Hemodynamical Flows: Modeling, Analysis and Simulation*. Birkhäuser Basel, 2008. DOI: 10.1007/978-3-7643-7806-6.

[3] G. Teschl. *Ordinary Differential Equations and Dynamical Systems*. Vol. 140. Graduate Studies in Mathematics. American Mathematical Society, 2012. DOI: 10.1090/gsm/140.

[4] A. R. Pries, D. Neuhaus, and P. Gaehtgens. "Blood viscosity in tube flow: dependence on diameter and hematocrit". In: *American Journal of Physiology-Heart and Circulatory Physiology* 263.6 (1992), H1770–H1778. DOI: 10.1152/ajpheart.1992.263.6.H1770.

[5] N. Kumar, A. Khader, R. Pai, P. A. Kyriacou, S. Khan, and P. Koteshwara. "Computational Fluid Dynamic Study on Effect of Carreau-Yasuda and Newtonian Blood Viscosity Models on Hemodynamic Parameters". In: *Journal of Computational Methods in Sciences and Engineering* 19.2 (2019), pp. 465–477. DOI: 10.3233/JCM-181004.

[6] H. Liu, L. Lan, J. Abrigo, H. L. Ip, Y. Soo, D. Zheng, K. S. Wong, D. Wang, L. Shi, T. W. Leung, and X. Leng. "Comparison of Newtonian and Non-Newtonian Fluid Models in Blood Flow Simulation in Patients With Intracranial Arterial Stenosis". In: *Frontiers in Physiology* 12 (2021), p. 718540. DOI: 10.3389/fphys.2021.718540.

[7] A. Quarteroni and L. Formaggia. "Mathematical Modelling and Numerical Simulation of the Cardiovascular System". In: *Handbook of Numerical Analysis*. Vol. 12. Used as broad cardiovascular-modeling background. Elsevier, 2004, pp. 3–127. DOI: 10.1016/S1570-8659(03)12001-7.

[8] Clay Mathematics Institute. *Navier–Stokes Equation*. Millennium Prize problem overview; modified 24 May 2023. Clay Mathematics Institute. May 27, 2022. URL: https://www.claymath.org/millennium/navier-stokes-equation/ (visited on 05/12/2026).

[9] L. Formaggia, D. Lamponi, and A. Quarteroni. "One-Dimensional Models for Blood Flow in Arteries". In: *Journal of Engineering Mathematics* 47.3–4 (2003), pp. 251–276. DOI: 10.1023/B:ENGI.0000007980.01347.29.

[10] Y. Shi, P. Lawford, and R. Hose. "Review of Zero-D and 1-D Models of Blood Flow in the Cardiovascular System". In: *BioMedical Engineering OnLine* 10.1 (2011), p. 33. DOI: 10.1186/1475-925X-10-33.

[11] A. Quarteroni and A. Veneziani. "Analysis of a Geometrical Multiscale Model Based on the Coupling of ODEs and PDEs for Blood Flow Simulations". In: *Multiscale Modeling & Simulation* 1.2 (2003), pp. 173–195. DOI: 10.1137/S1540345902408482.

[12] L. Formaggia, A. Moura, and F. Nobile. "On the Stability of the Coupling of 3D and 1D Fluid-Structure Interaction Models for Blood Flow Simulations". In: *ESAIM: Mathematical Modelling and Numerical Analysis* 41.4 (2007), pp. 743–769. DOI: 10.1051/m2an:2007039.

[13] S. Canic, S. Guo, Y. Wang, X. Yue, and H. Zheng. "Extended one-dimensional reduced model for blood flow within a stenotic artery". In: *arXiv preprint arXiv:2409.16262* (2024). arXiv:2409.16262v1. DOI: 10.48550/arXiv.2409.16262. URL: https://arxiv.org/abs/2409.16262.

[14] R. J. LeVeque. *Finite Volume Methods for Hyperbolic Problems*. Cambridge University Press, 2002. DOI: 10.1017/CBO9780511791253.

[15] E. Boileau, P. Nithiarasu, P. J. Blanco, L. O. Muller, F. E. Fossan, L. R. Hellevik, W. P. Donders, W. Huberts, M. Willemet, and J. Alastruey. "A Benchmark Study of Numerical Schemes for One-Dimensional Arterial Blood Flow Modelling". In: *International Journal for Numerical Methods in Biomedical Engineering* 31.10 (2015), e02732. DOI: 10.1002/cnm.2732.

[16] F. Derimay, G. Finet, and G. Rioufol. "Coronary Artery Stenosis Prediction Does Not Mean Coronary Artery Stenosis Obstruction". In: *European Heart Journal* 42.42 (2021), p. 4401. DOI: 10.1093/eurheartj/ehab332.

[17] J. Escaned and J. Davies, eds. *Physiological Assessment of Coronary Stenoses and the Microcirculation*. Springer London, 2017. DOI: 10.1007/978-1-4471-5245-3.

[18] K. Shaikh, P. R. Lozano, S. Evangelou, E.-H. Wu, N. S. Nurmohamed, N. Madan, D. Verghese, C. Shekar, A. Waheed, S. Siddiqui, M. R. Kolossváry, S. Almeida, T. Coombes, D. Suchá, S. J. Trivedi, and A. R. Ihdayhid. "CT derived fractional flow reserve: Part 1 – Comprehensive review of methodologies". In: *Journal of Cardiovascular Computed Tomography* 19.4 (2025). SCCT FiRST (Fellow and Resident Leaders of SCCT) Committee, pp. 390–396. DOI: 10.1016/j.jcct.2025.07.001.

[19] I. J. van den Hoogen, J. Schultz, J. H. Kuneman, M. A. de Graaf, V. Kamperidis, A. Broersen, J. W. Jukema, A. Sakellarios, S. Nikopoulos, S. Kyriakidis, K. K. Naka, L. Michalis, D. I. Fotiadis, T. Maaniitty, A. Saraste, J. J. Bax, and J. Knuuti. "Detailed Behaviour of Endothelial Wall Shear Stress Across Coronary Lesions From Non-Invasive Imaging With Coronary Computed Tomography Angiography". In: *European Heart Journal - Cardiovascular Imaging* 23.12 (2022), pp. 1708–1716. DOI: 10.1093/ehjci/jeac095.

[20] N. H. Pijls, B. De Bruyne, K. Peels, P. H. Van Der Voort, H. J. Bonnier, J. Bartunek, and J. J. Koolen. "Measurement of fractional flow reserve to assess the functional severity of coronary-artery stenoses". In: *The New England Journal of Medicine* 334.26 (1996), pp. 1703–1708. DOI: 10.1056/NEJM199606273342604.

[21] P. A. L. Tonino, W. F. Fearon, B. De Bruyne, K. G. Oldroyd, M. A. Leesar, P. N. Ver Lee, P. A. Maccarthy, M. Van't Veer, and N. H. J. Pijls. "Angiographic versus functional severity of coronary artery stenoses in the FAME study fractional flow reserve versus angiography in multivessel evaluation". In: *Journal of the American College of Cardiology* 55.25 (2010), pp. 2816–2821. DOI: 10.1016/j.jacc.2009.11.096.

[22] B. L. Nørgaard, J. Leipsic, S. Gaur, S. Seneviratne, B. S. Ko, H. Ito, J. M. Jensen, L. Mauri, B. De Bruyne, H. Bezerra, K. Osawa, M. Marwan, C. Naber, A. Erglis, S.-J. Park, E. H. Christiansen, A. Kaltoft, J. F. Lassen, H. E. Bøtker, S. Achenbach, and NXT Trial Study Group. "Diagnostic performance of noninvasive fractional flow reserve derived from coronary computed tomography angiography in suspected coronary artery disease: the NXT trial". In: *Journal of the American College of Cardiology* 63.12 (2014), pp. 1145–1155. DOI: 10.1016/j.jacc.2013.11.043.

[23] L. John, P. Pustějovská, and O. Steinbach. "On the influence of the wall shear stress vector form on hemodynamic indicators". In: *Computing and Visualization in Science* 18.4–5 (2017), pp. 113–122. DOI: 10.1007/s00791-017-0277-7.